ASUS ZB570TL Maintenance Manual

Technical documents

file name : ASUS ZB570TL Maintenance Manual
version : V1.0

Fake :
Review :
Will sign :

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RF Signal Generator

DC power supply

Oscilloscope

Hot air gun

Electric iron

Spectrum Analyzer

electric screw driver

UP-1024 High speed programmer

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Disassembly tool

multimeter

Anti-static gloves

Lead-free solder wire

Anti-static cloth

alcohol bottle

Pen brush

Anti-static brush

Anti-static fixture

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ID Exterior

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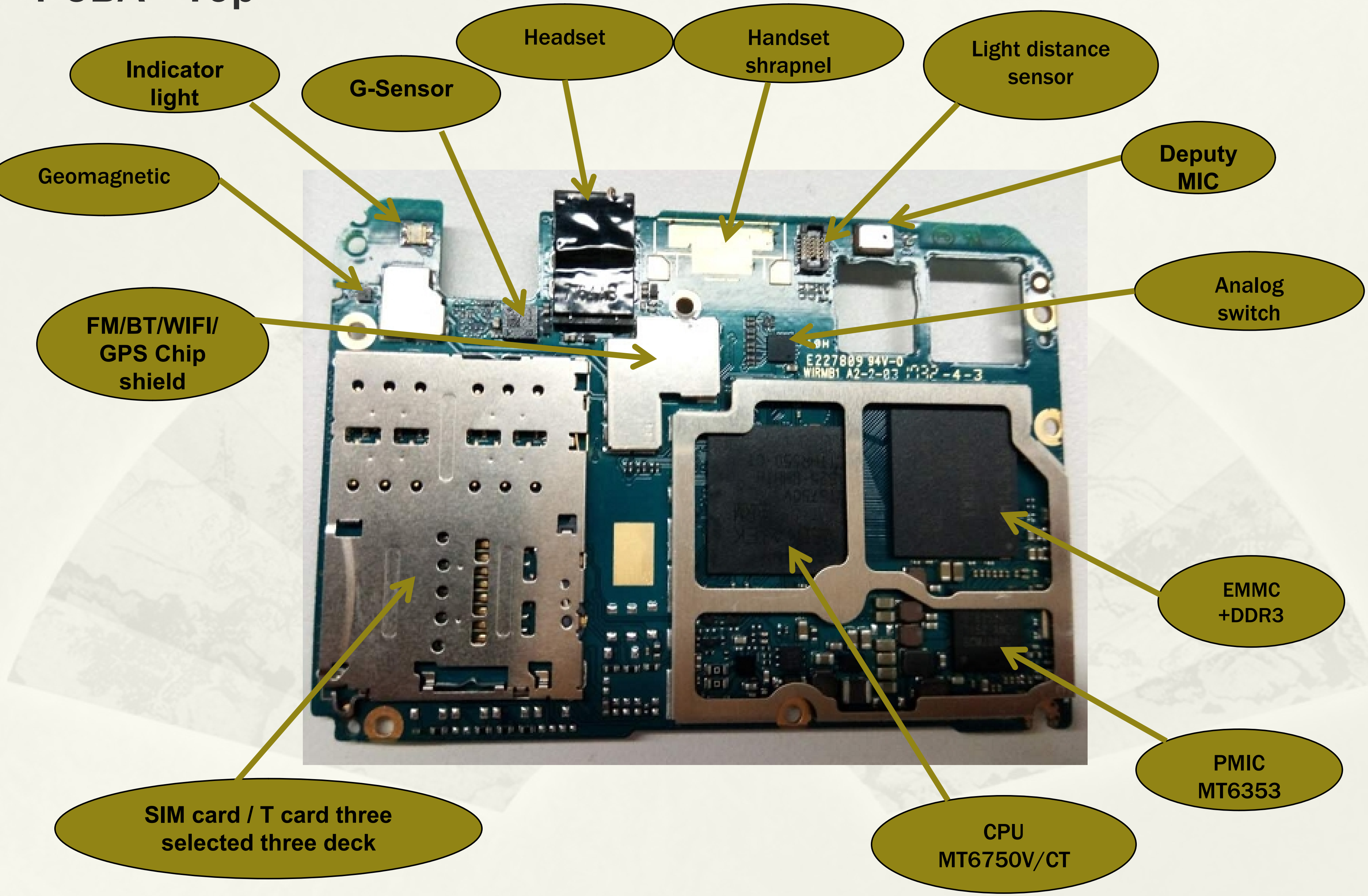
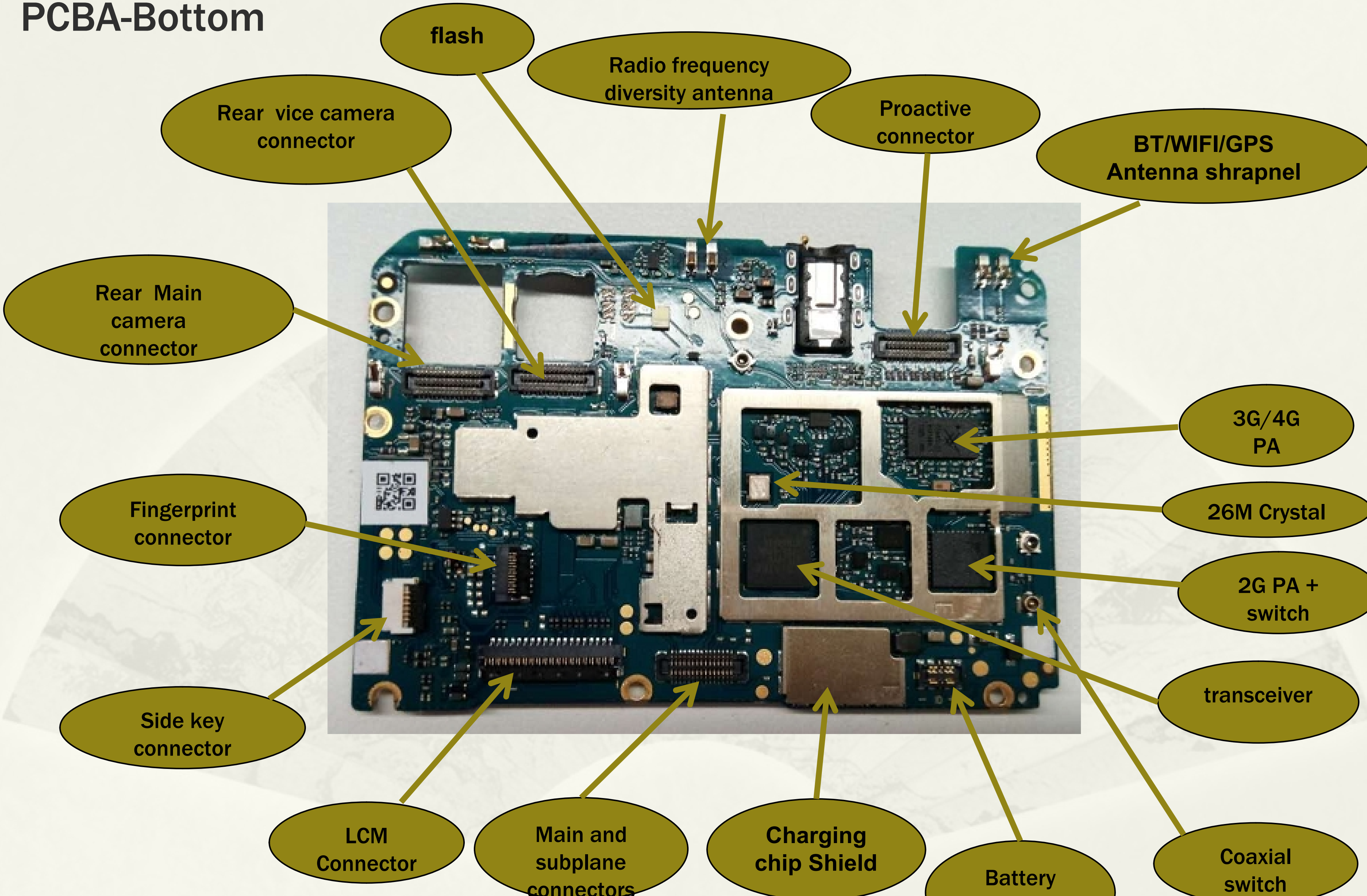
PCBA –Top


Indicator light
G-Sensor
Headset
Handset shrapnel
Light distance sensor
Deputy MIC
Geomagnetic
Analog switch
FM/BT/WIFI/ GPS Chip shield
EMMC +DDR3
PMIC MT6353
SIM card / T card three selected three deck
CPU MT6750V/CT

PCBA-Bottom


flash
Radio frequency diversity antenna
Rear vice camera connector
Proactive connector
BT/WIFI/GPS Antenna shrapnel
Rear Main camera connector
3G/4G PA
26M Crystal
Fingerprint connector
2G PA + switch
transceiver
Side key connector
LCM Connector
Main and subplane connectors
Charging chip Shield
Battery connector
Coaxial switch

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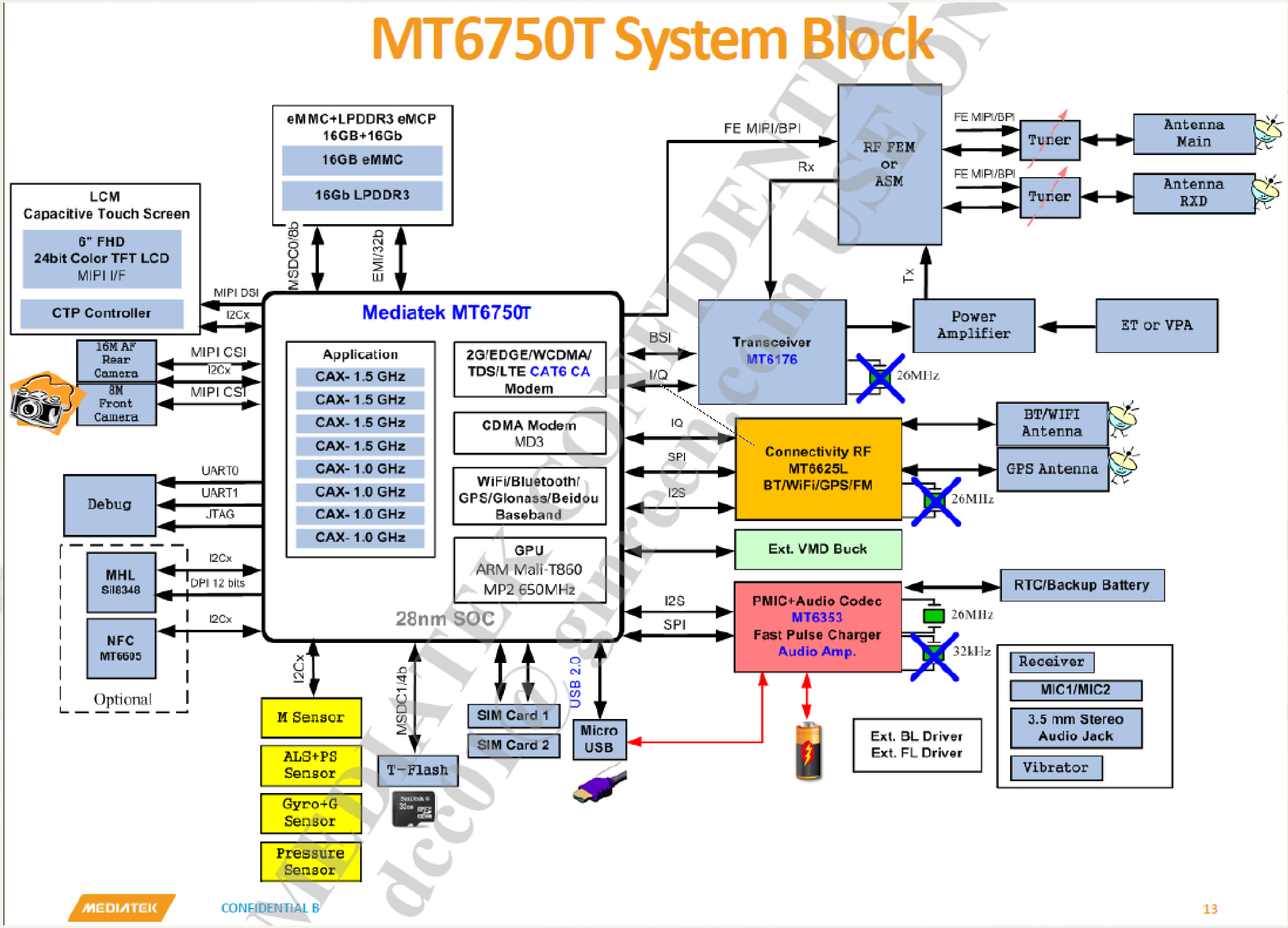
System Block Diagram



Button switch machine timing

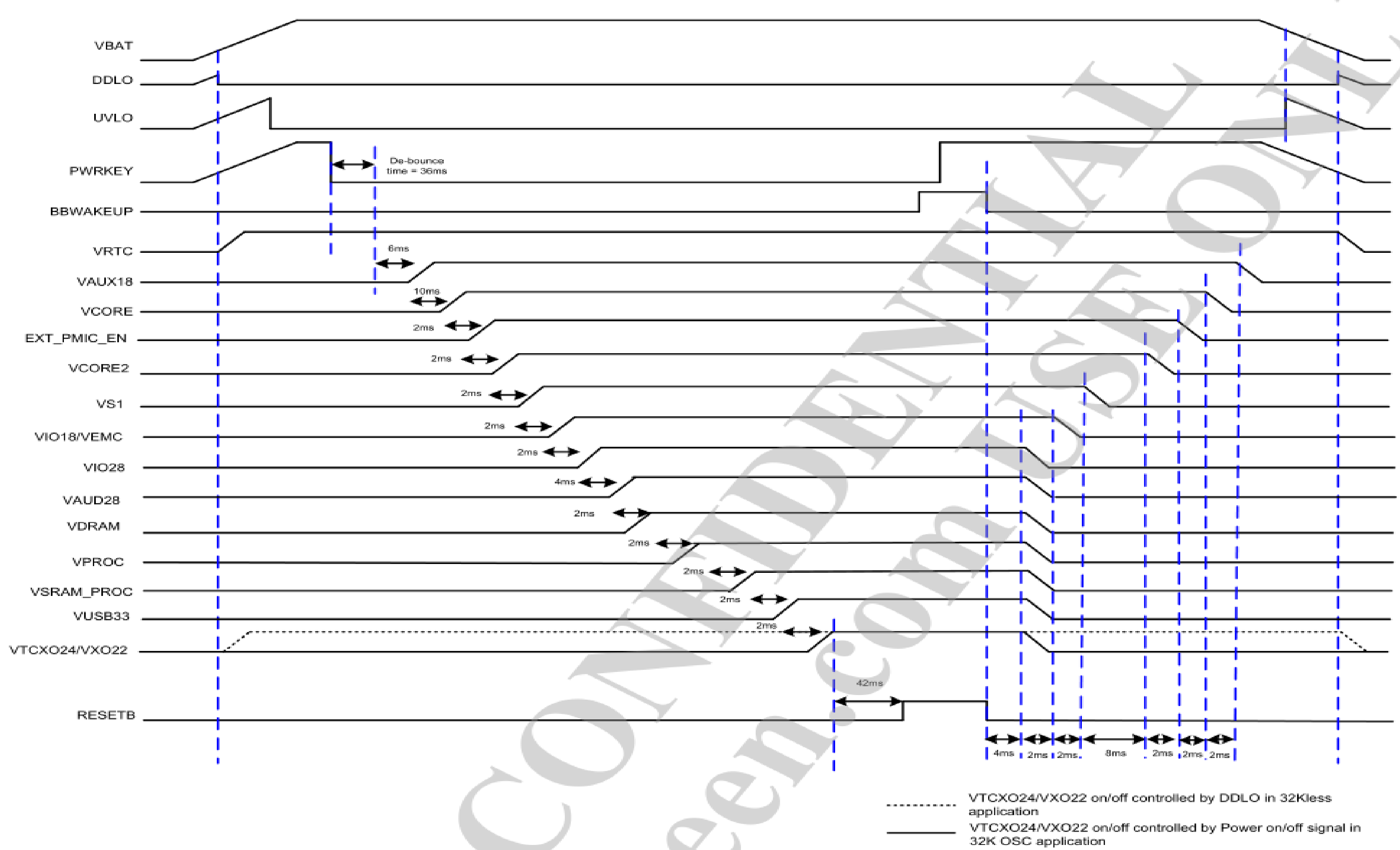


Figure 3-2. Power-on/off control sequence with XTAL by pressing PWRKEY

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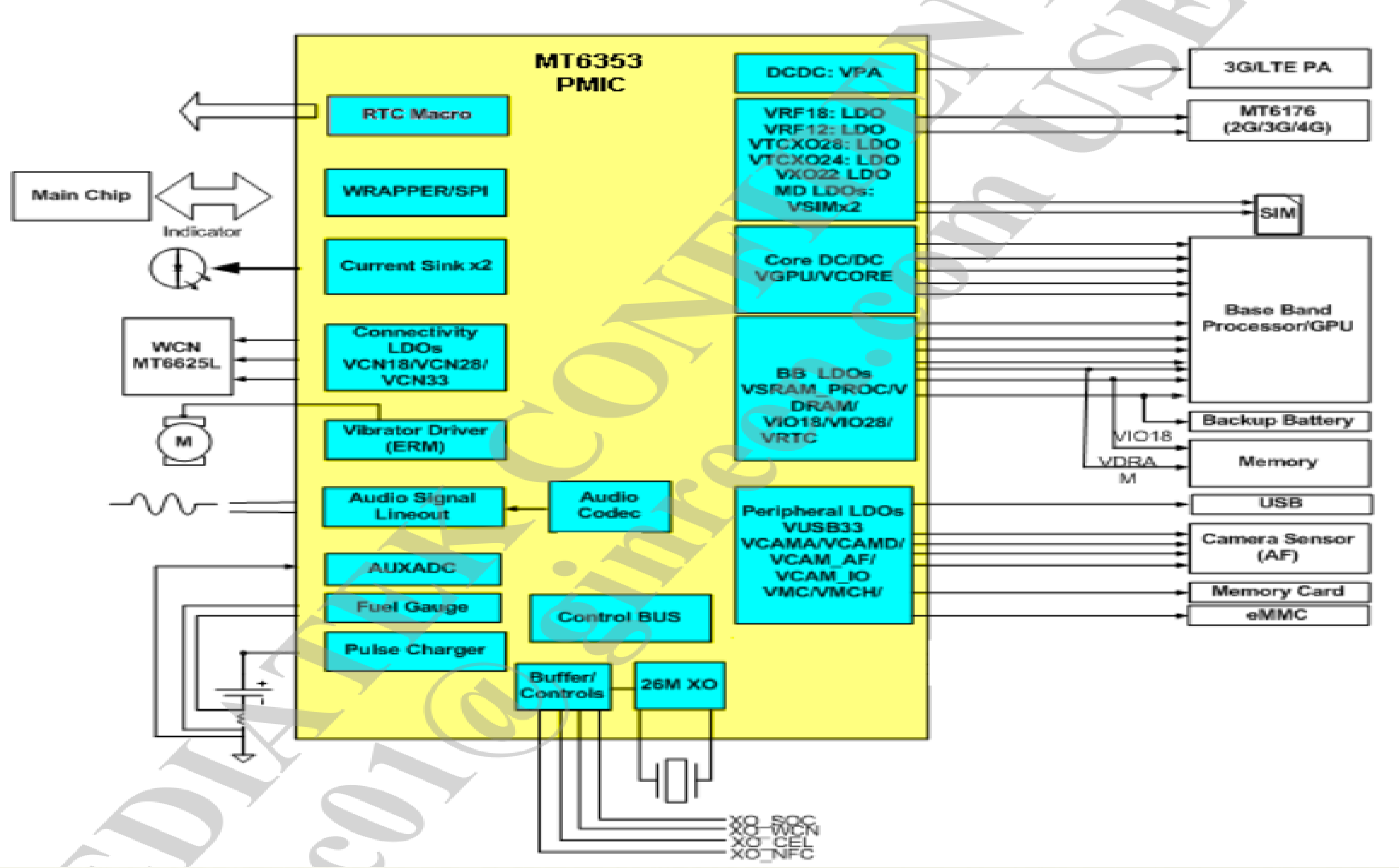
I2C address

器件类型	IC型号	IC厂家	I2C设备地址	I2C设备写地址	I2C设备读地址	备注		
CAMERA	HI846	Hynix	0x23	0x46	0x47	8M	8M前摄（一供）	I2C1
	OV8856	OV	0x36	0x6C	0x6D	8M	8M前摄（二供）	
	HI1332	Hynix	0x20	0x40	0x0X41	13M	13M前摄（一供）	
	OV13858	OV	0x36	0x6C	0x6D	13M	13M前摄（二供）	
	S5K3P3	三星	0x2D	0x5A	0x5B	16M	16M前摄（一供）	
	OV16880	OV	0x36	0x6C	0x6D	16M	16M前摄（二供）	
	OV8856	OV	0x10	0x20	0x21	8M	8M广角	I2C2
	S5K3L8	三星	0x10	0x20	0x21	13M	13M后摄（一供）	
	OV13855	OV	0x10	0x20	0x21	13M	13M后摄（二供）	
	S5K3P3	三星	0x10	0x20	0x21	16M	16M后摄（一供）	
	OV16880	OV	0x10	0x20	0x21	16M	16M后摄（二供）	
充电芯片	TI	BQ25896RTWR	0x6B	0xD6	0xD7	充电芯片		I2C1
CTP	MSG2836A	Mstar	0x26	0x4C	0x4D	CTP		I2C0
LCM驱动电压偏压芯片	NT50358CG/A	NOVATEK	0x3E	0x7C	0x7D	LCM驱动电压偏压芯片	一供	I2C0
	LP3101SCCPR	LEPOWER	0x3E	0x7C	0x7D		二供	
	TPS65132A0YFFR	TI	0x3E	0x7C	0x7D		一供	
G-Senser	LSM6DS33	ST	0x6A	0xD4	0xD5	G-Senser		I2C1
地磁传感器	AF8133J	VTC	0x1E	0x3C	0x3D	Pin A0接GND		I2C1
			0x1C	0x38	0x39	Pin A0拉高		
三合一传感器	PA22A00001	TXC	0x1E	0x3C	0x3D	三合一传感器		I2C2
闪光灯芯片	AW3648CSR	AWINIC	0x63	0xC6	0xC7	闪光灯芯片		I2C1
	TLV61310YFFR	TI	0x3E	0x7C	0x7D			I2C1

Power management



Buck Power management

BUCK name	Default voltage (V)	Vout (Volt)	Voltage step (mV)	Imax (mA)	Default on (Y/N)	Application
VCORE	0.9	0.6 ~ 1.31	6.25	3,000	Y	Digital core always
VCORE2	1	0.6 ~ 1.31	6.25	3,500	Y	MODEM
VPROC	1	0.6 ~ 1.31	6.25	5,000	Y	PROCESSOR
VS1	2	1.95 ~ 2.20	12.5	1,900	Y	SYS LDOs power
VPA	0.5	0.5~3.4	50	800	N	3G/LTE PA

LDO Power management （1）

Table 3-2. LDO types and brief specifications

Type	LDO name	Input power domain	Default Voltage	Vout (Volt)	Imax (mA)	Default on (Y/N)	Application
ALDO	VTCXO28	VBAT_LDO1	2.8	2.8	40mA	N	RFFE
ALDO	VTCXO24	VBAT_LDO2	2.375	2.375	40mA	Y	DCXO
ALDO	VXO22	VBAT_LDO2	2.2	2.2	20mA	Y	DCXO
SLDO	VRF18	AVDD20_LDO5	1.825	1.825	400mA	N	RF
SLDO	VRF12	AVDD20_LDO5	1.2	1.2	150mA	N	RF
DLDO	VCN33	VBAT_LDO2	3.3	3.3/3.4/3.5/3.6	350mA	N	WCN
ALDO	VCN28	VBAT_LDO1	2.8	2.8	40mA	N	WCN
SLDO	VCN18	AVDD20_LDO2	1.8	1.8	150mA	N	WCN
ALDO	VCAMA	VBAT_LDO1	2.8	2.5/ 2.8	200mA	N	CAMERA
SLDO	VCAMIO	AVDD20_LDO2	1.8	1.8	200mA	N	CAMERA

LDO Power management （2）

Type	LDO name	Input power domain	Default Voltage	Vout (Volt)	Imax (mA)	Default on (Y/N)	Application
SLDO	VCAMD	AVDD20_LDO2	1.3	0.9/1.0/1.1/1.22/1.3/1.5/1.8	<=1.5V: 500mA 1.8V: 300mA	N	CAMERA
ALDO	VAUX18	VBAT_LDO1	1.8	1.8	40mA	Y	AuxADC
ALDO	VAUD28	VBAT_LDO1	2.8	2.8	40mA	Y	Audio
SLDO	VSRAM_PROC	AVDD20_LDO4	1.1	0.6~1.31	400mA	Y	SRAM_PROC
SLDO	VDRAM	AVDD20_LDO1	1.24	1.24	1,000mA	Y	DRAM
DLDO	VSIM1	VBAT_LDO4	1.8	1.7/1.8/1.86/2.76/3.0/3.1	50mA	N	SIM1 card
DLDO	VSIM2	VBAT_LDO4	1.8	1.7/1.8/1.86/2.76/3.0/3.1	50mA	N	SIM2 card

LDO Power management （3）

Type	LDO name	Input power domain	Default Voltage	Vout (Volt)	Imax (mA)	Default on (Y/N)	Application
DLDO	VLDO28	VBAT_LDO4	2.8	2.8	300mA	N	Touch Panel and CAMERA _AF
DLDO	VIO28	VBAT_LDO4	2.8	2.8	200mA	Y	IO and Sensor
DLDO	VMC	VBAT_LDO5	3	1.8/2.9/3.0/ 3.3	200mA	N	MSDC
DLDO	VMCH	VBAT_LDO5	3	2.9/3.0/3.3	800mA	N	SD Card
DLDO	VEMC33	VBAT_LDO3	3	2.9/3.0/3.3	800mA	Y	EMMC and UFS
DLDO	VIBR	VBAT_LDO4	2.8	1.2/1.3/1.5/1 .8/2.5/2.8/3 .0/3.3	100mA	N	Vibrator
DLDO	VUSB33	VBAT_LDO2	3.3	3.3	50mA	Y	USB
SLDO	VIO18	AVDD20_LDO3	1.8	1.8	600mA	Y	IO and EMMC
VRTC28	VRTC28	VBAT_LDO1	2.8	2.8	2mA	Y	RTC macro
VDIG18	DVDD18_DIG	VBAT_LDO2	1.8	1.8	20mA	Y	PMIC digital
Power Switch	TREF	VBAT_LDO1	1.8	1.8	1mA	N	Battery Interface

Charge switch machine timing

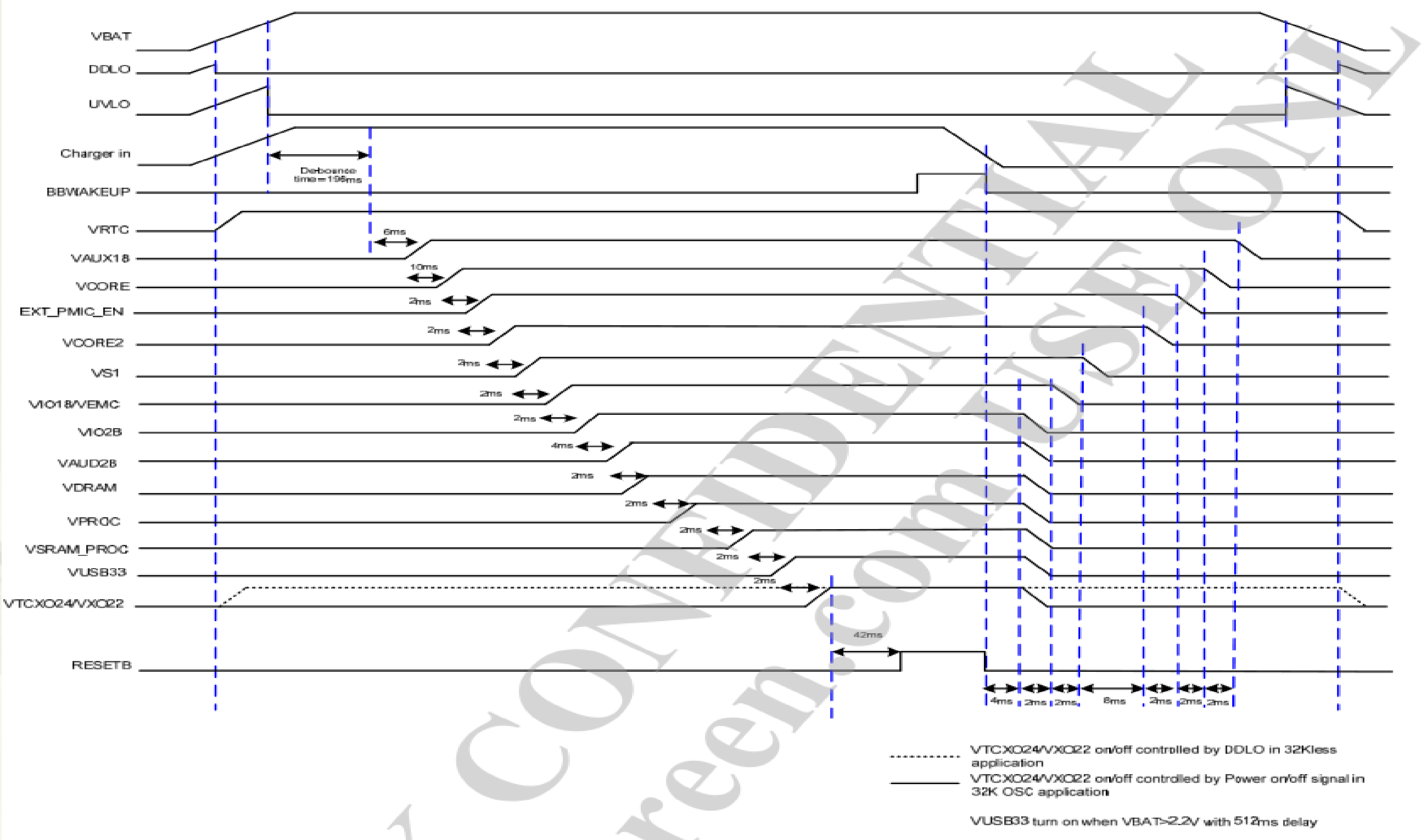


Figure 3-3. Power-on/off control sequence with XTAL by charger plug in

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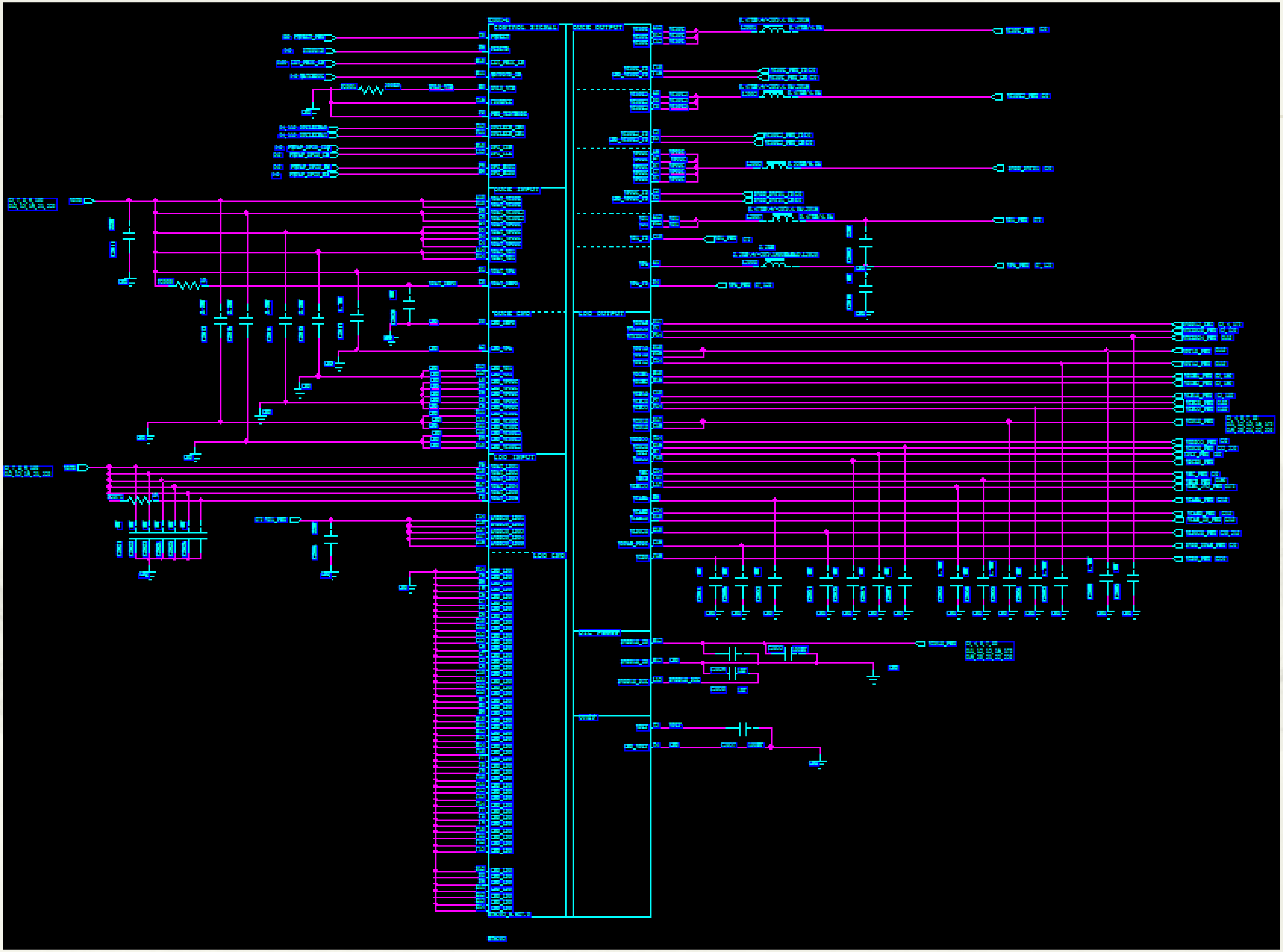
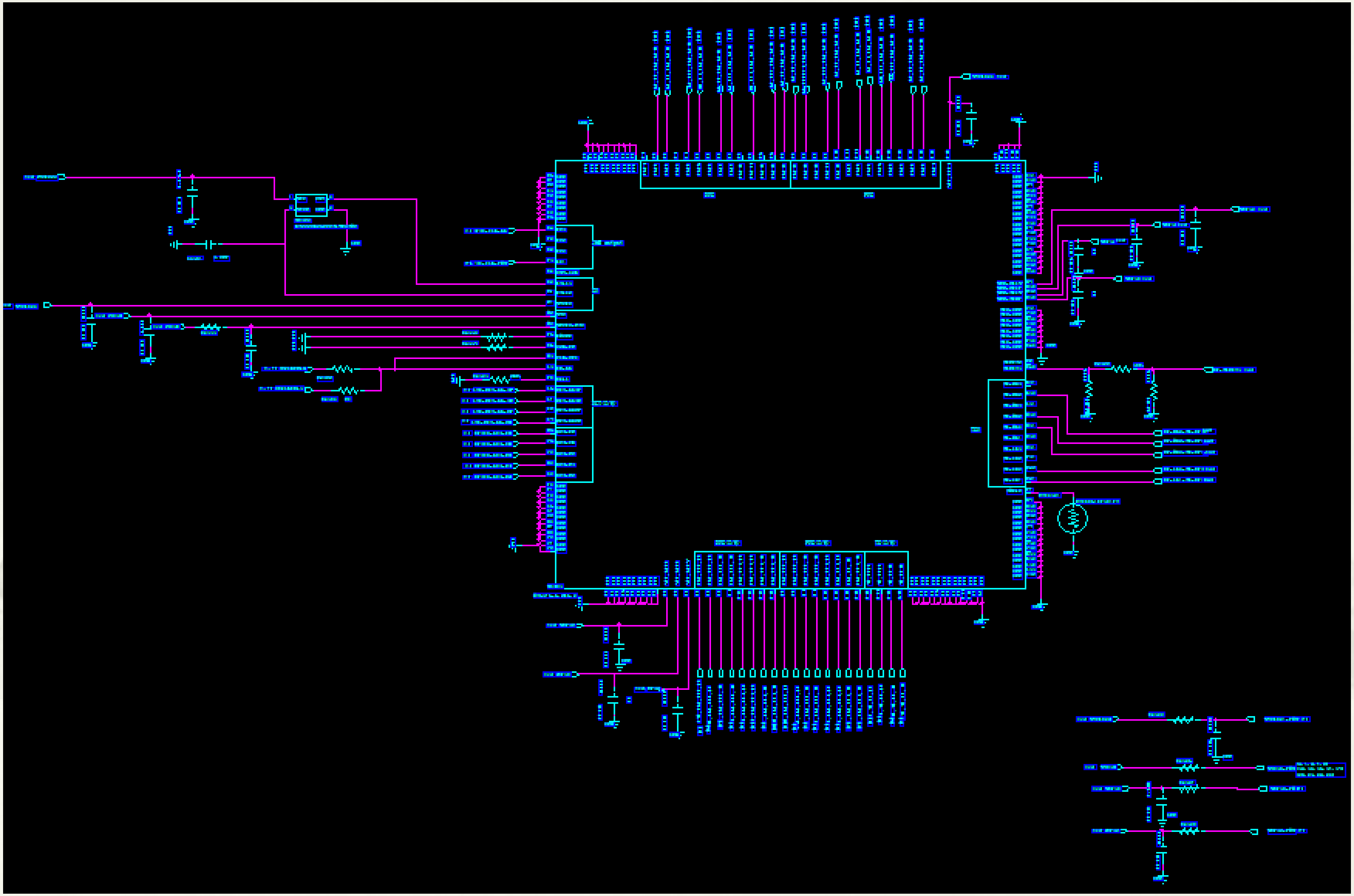
Power supply：MT6353 U2001

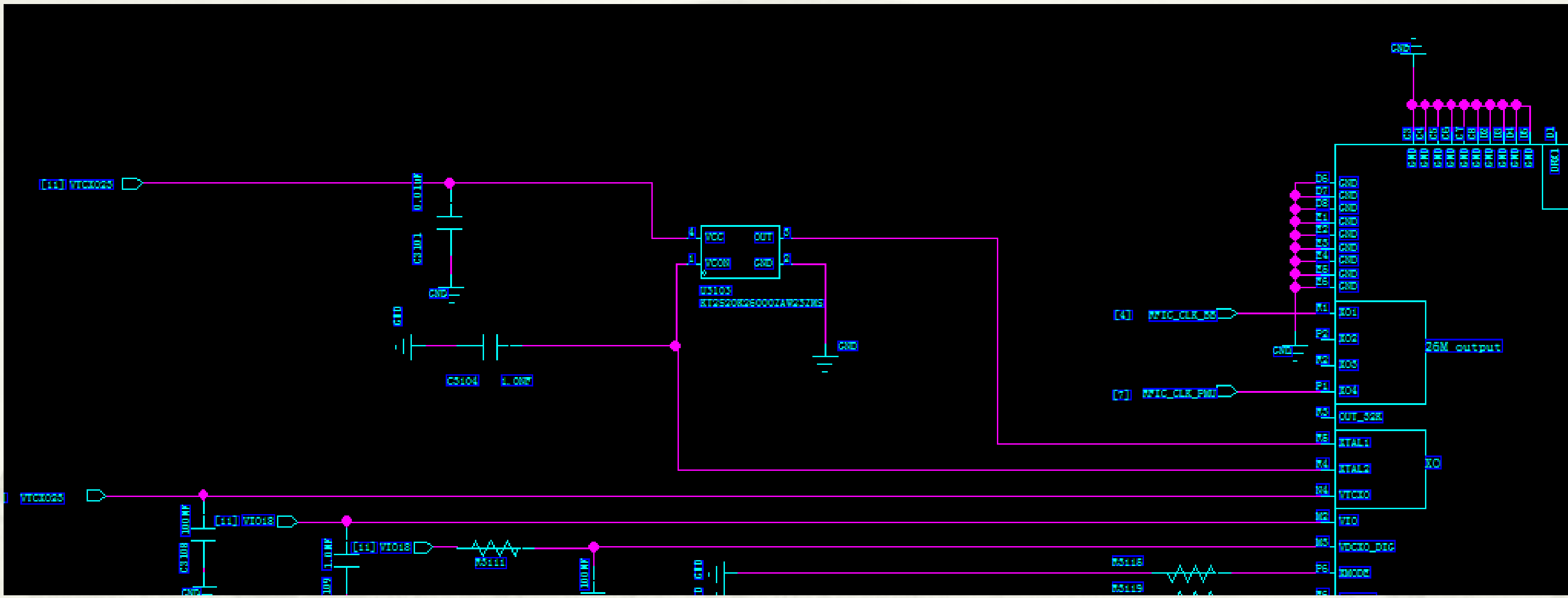


Transceiver MT6176 U3101

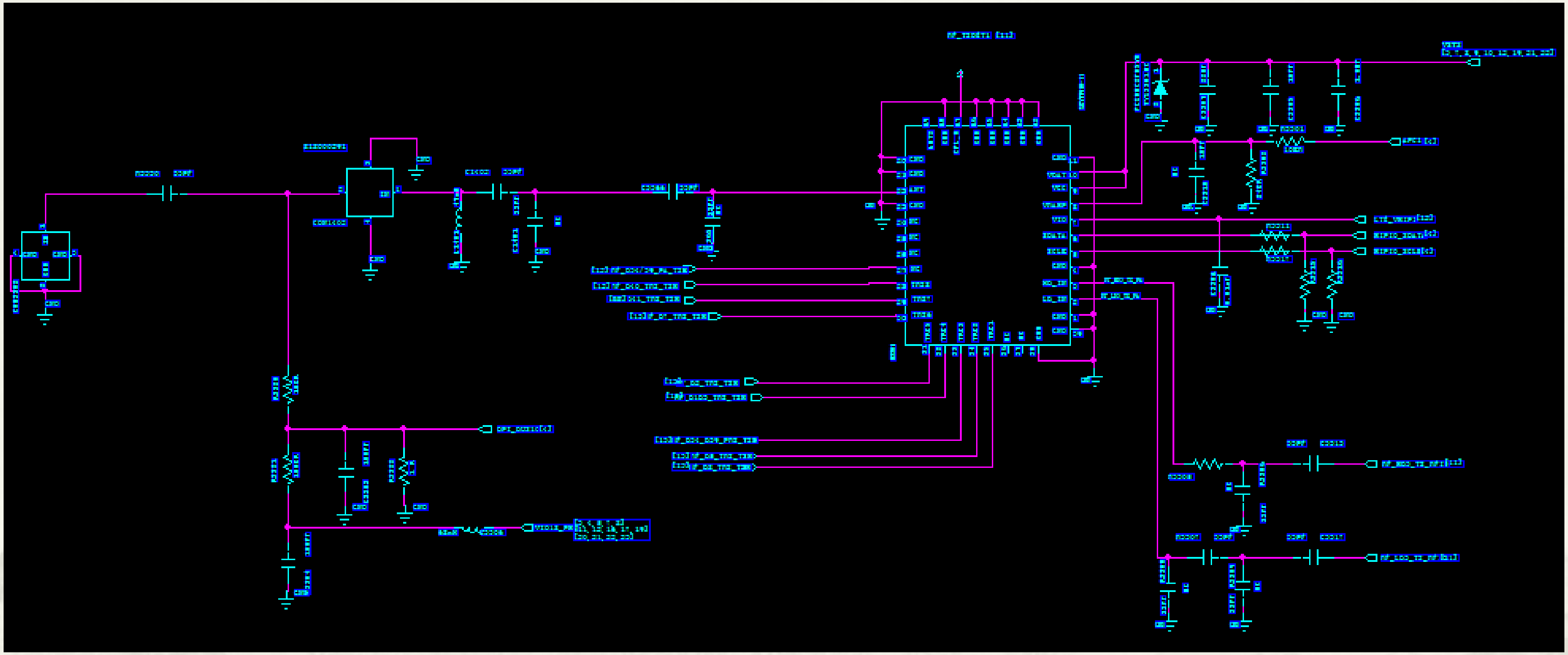


26MHz Crystal section U3103

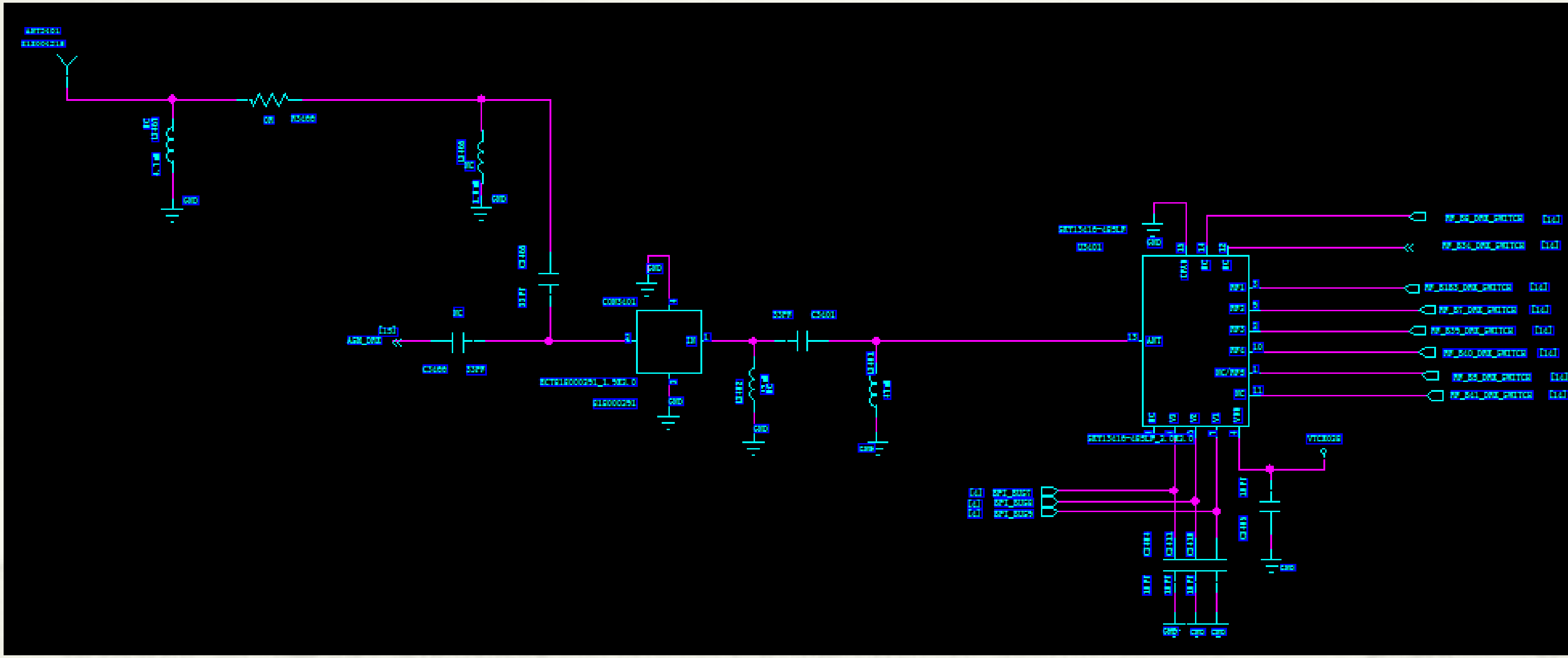
The output wave of Crystal looks like sine wave with amplitude 1V, frequency 26MHz



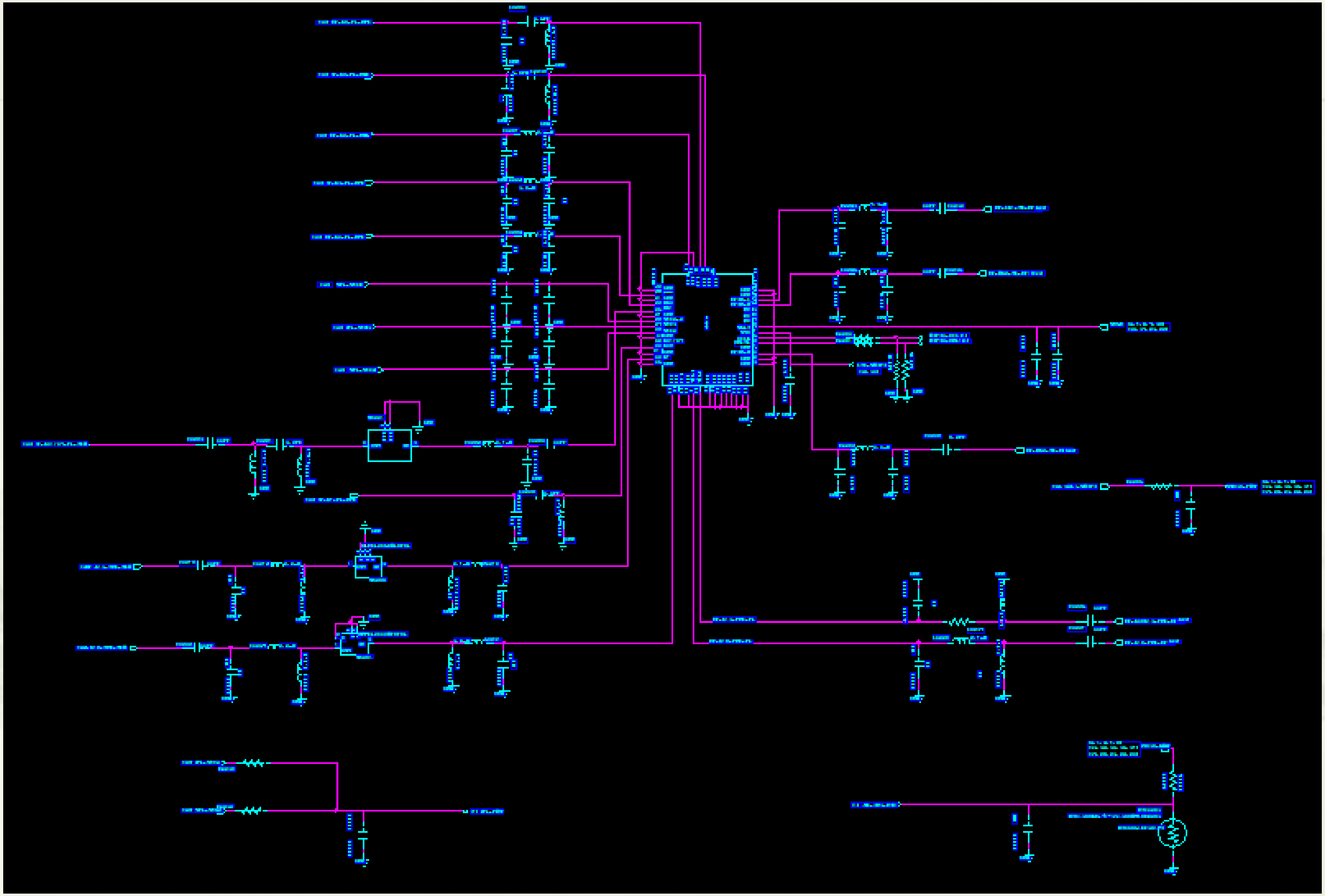
SP16T U3301



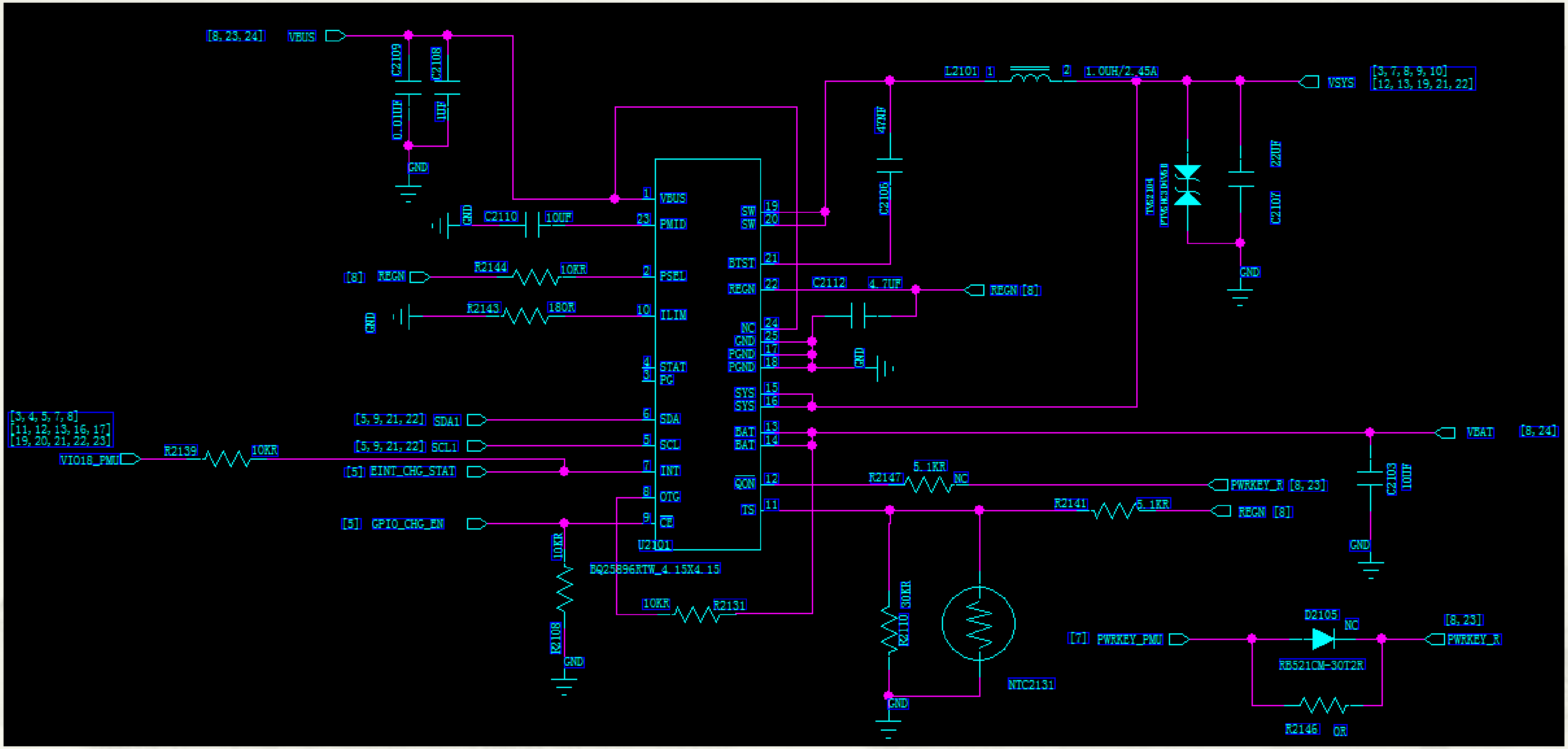
SP6T U3401



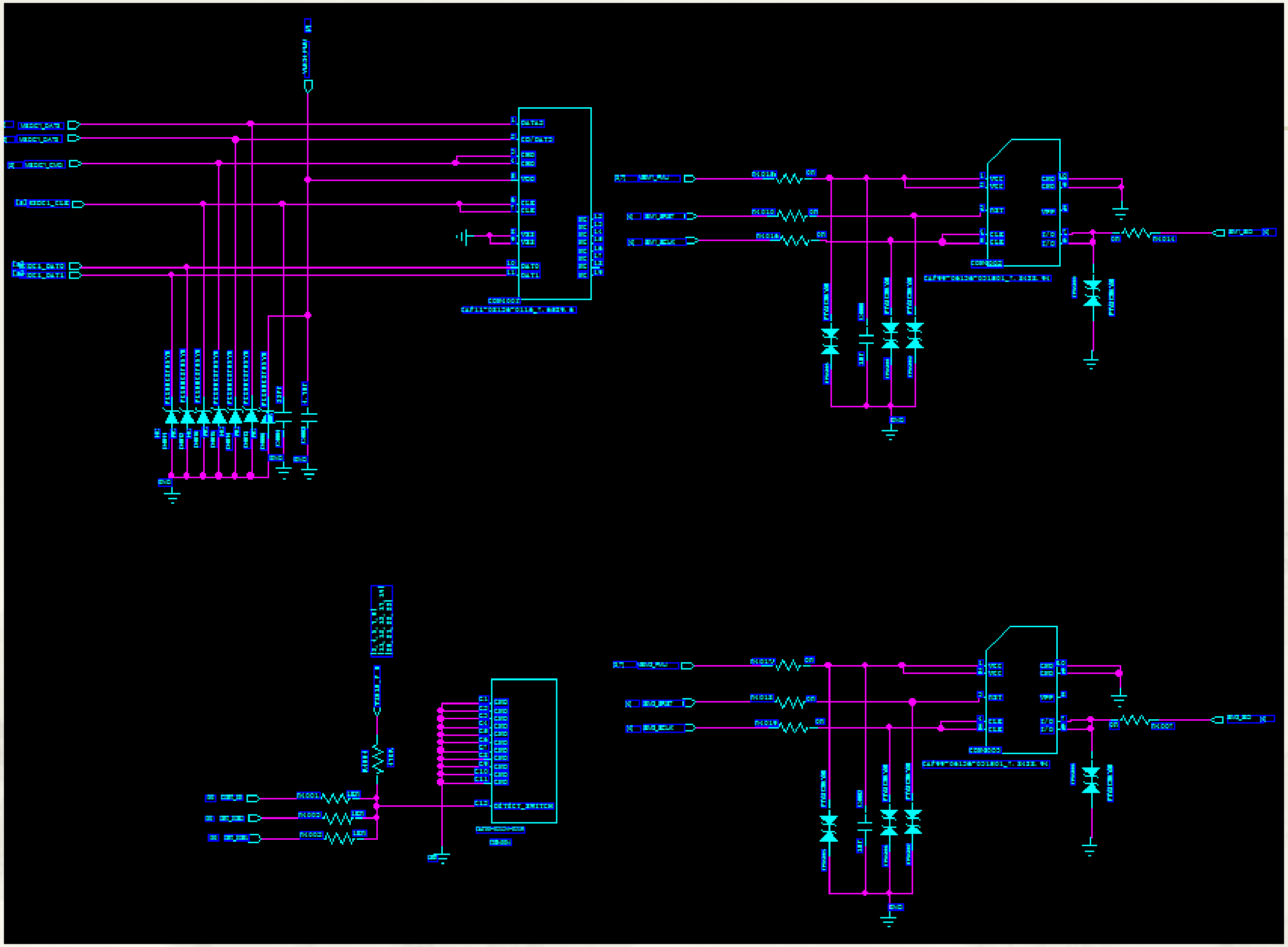
MMPA U3201



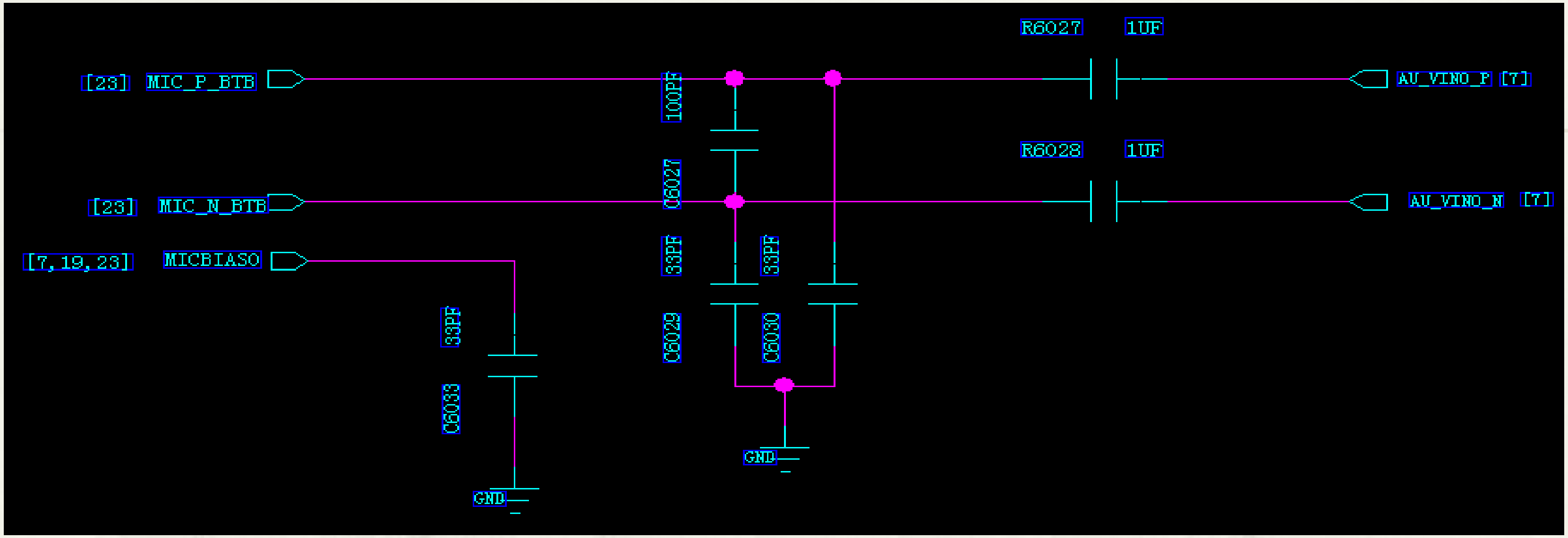
Charge management BQ25896 U2101



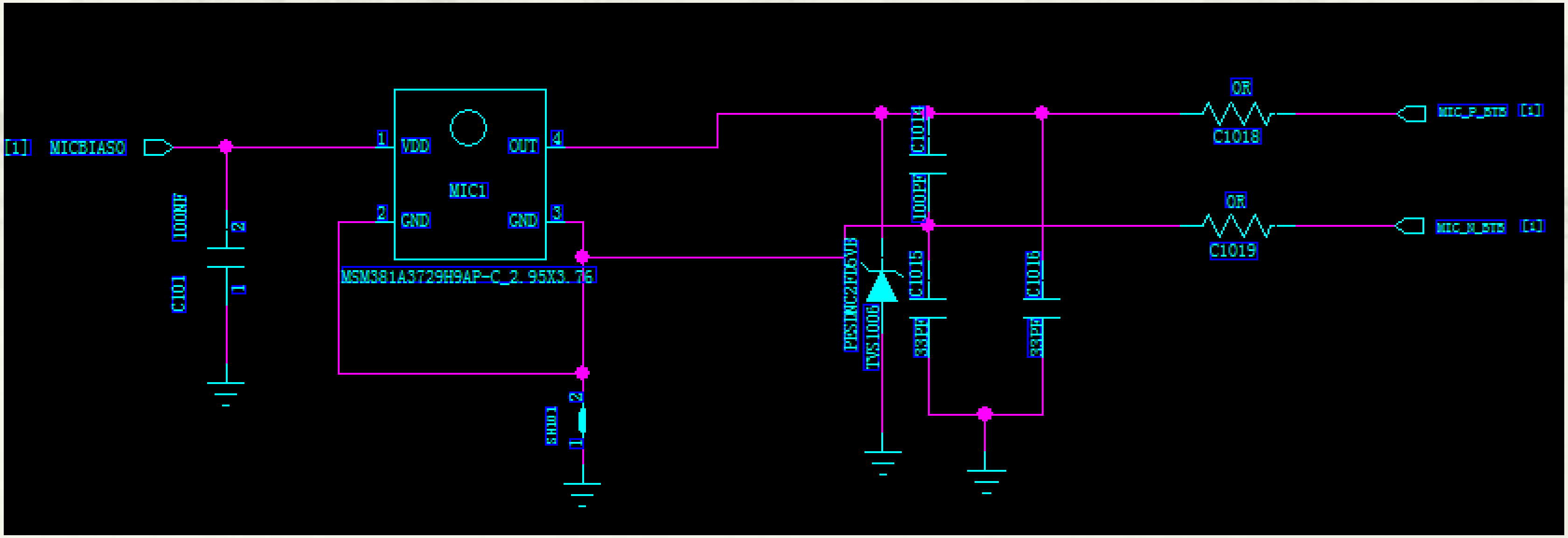
SIM & TF socket U4001



Main MIC

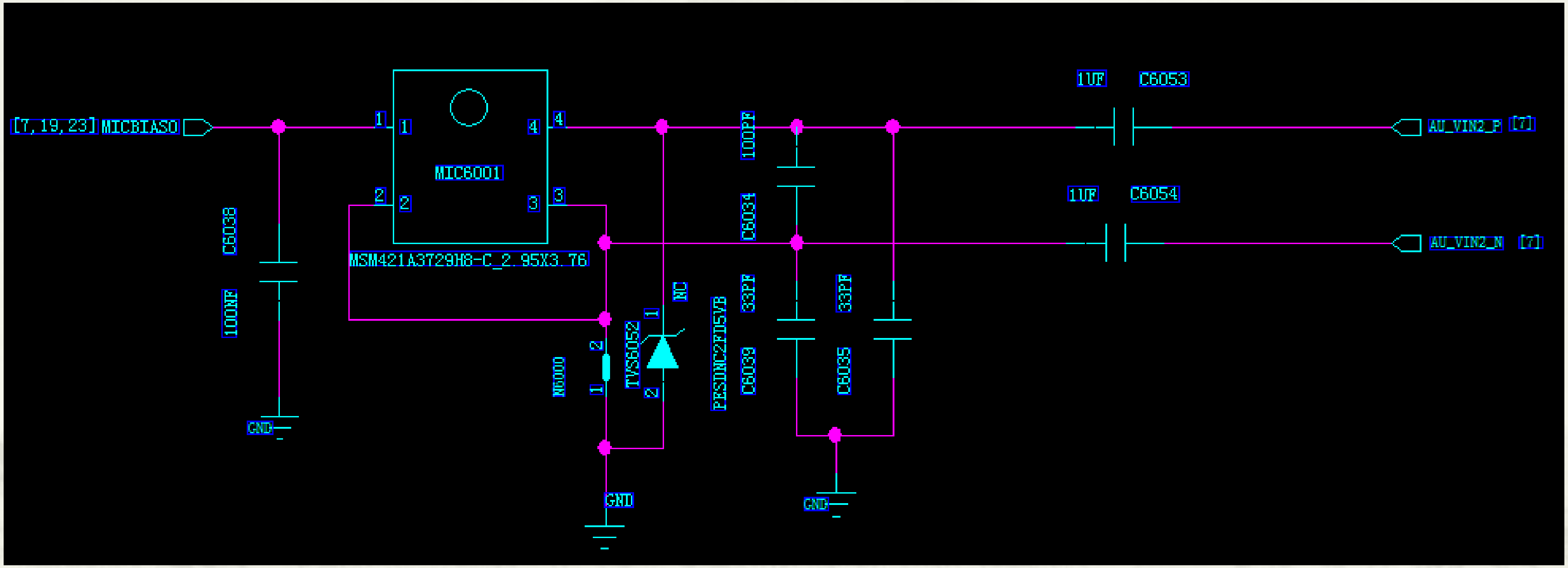


R6027
1UF
[23] MIC_P_BTB
AU_VINO_P [7]
R6028
1UF
[23] MIC_N_BTB
AU_VINO_N [7]
100PF
C6027
[7,19,23] MICBIASO
33PF
C6029
33PF
C6030
33PF
C6033
GND
GND



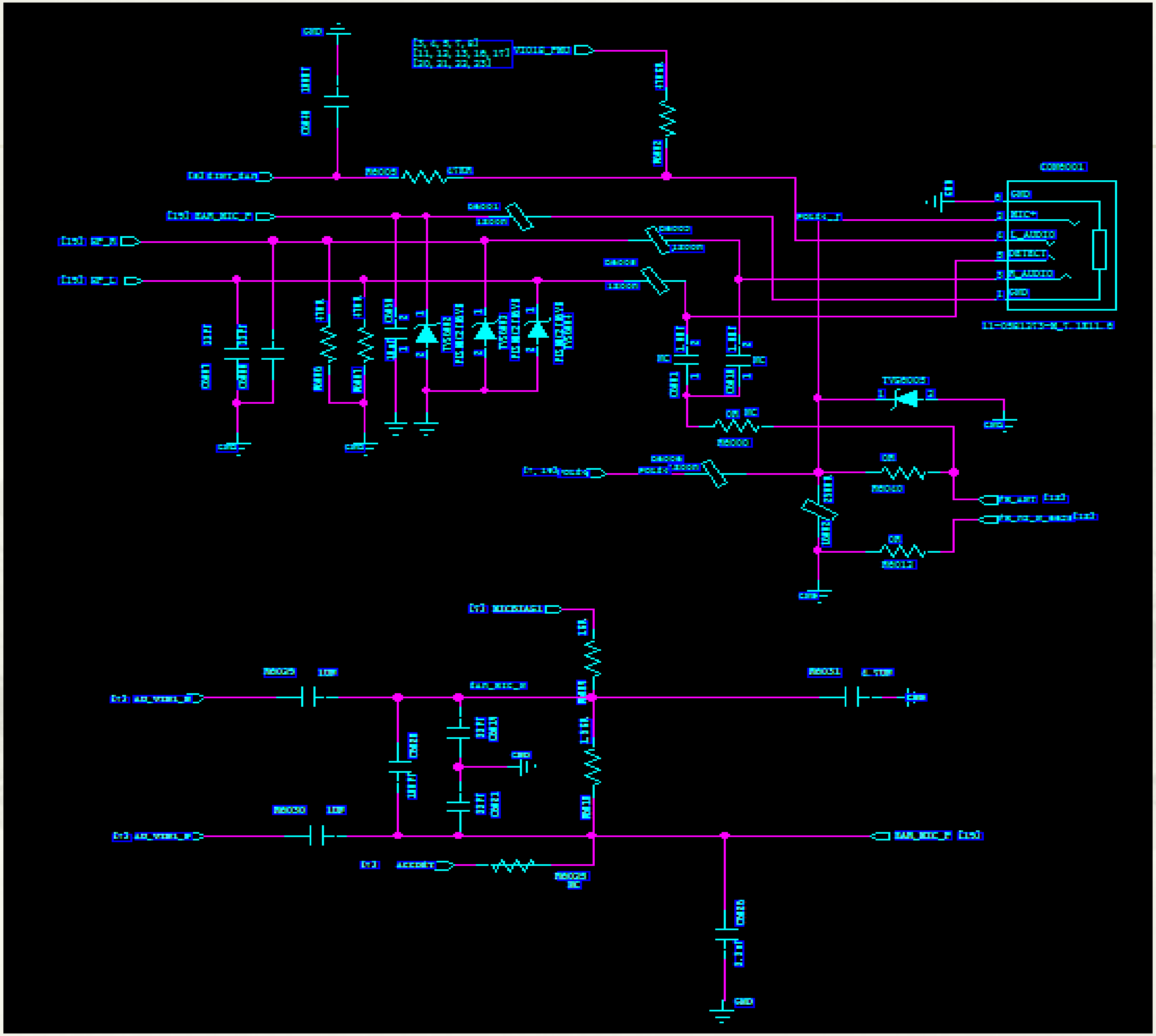
[1] MICBIASO
100NF
C101
VDD
OUT
MIC1
GND
GND
MSM381A3729H9AP-C_2.95X3.76
C1014
100PF
C1015
33PF
C1016
33PF
TVS1006
OR
C1018
OR
C1019
MIC_P_BTB [1]
MIC_N_BTB [1]

Second MIC

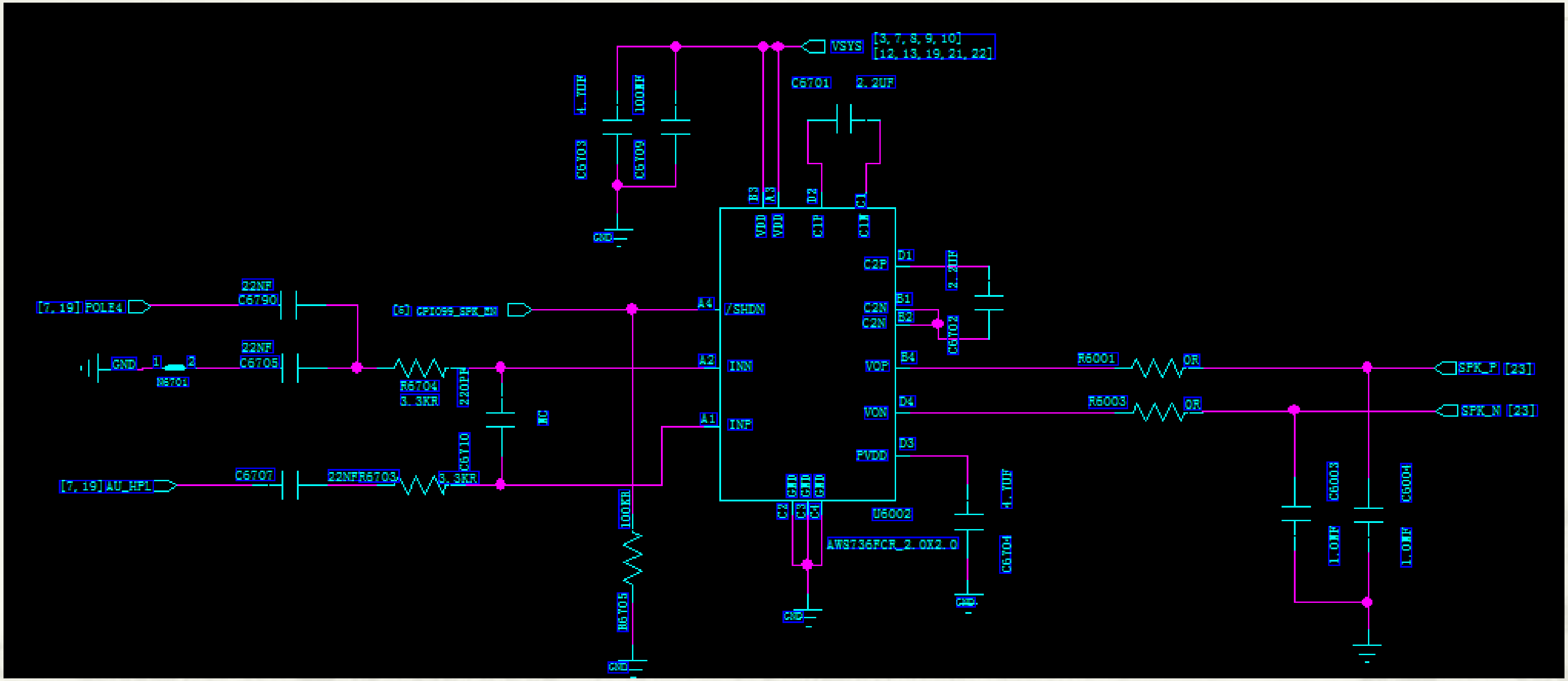


1UF
C6053
[7,19,23] MICBIAS0
1
1
4
4
MIC6001
AU_VIN2_P [7]
100PF
1UF
C6054
2
2
3
3
C6034
AU_VIN2_N [7]
C6038
MSM421A3729H8-C_2.95X3.76
NC
33PF
33PF
100NF
1
TVS6052
PESDNC2FD5VB
2
N6000
1
2
C6039
C6035
GND
GND
GND

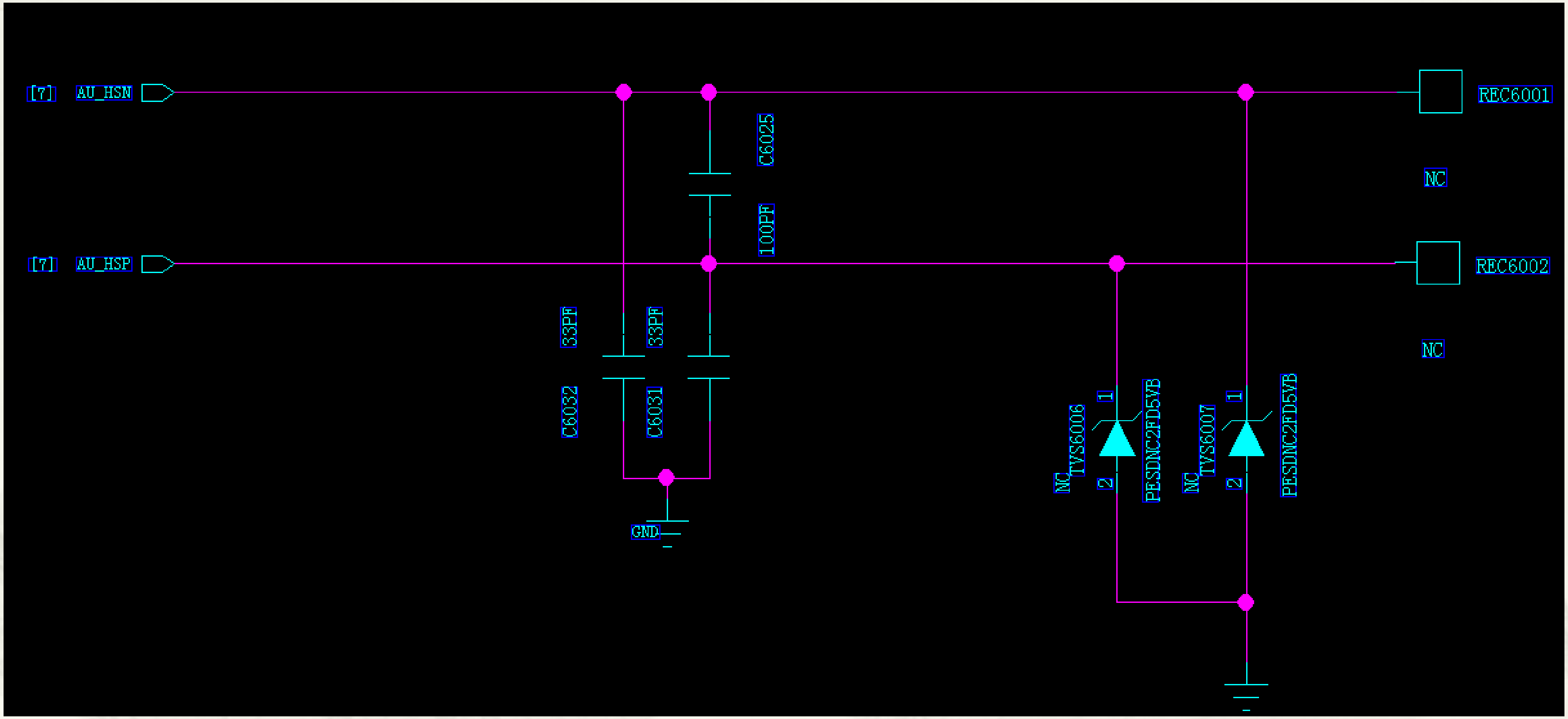
Headset Jack section Q6001



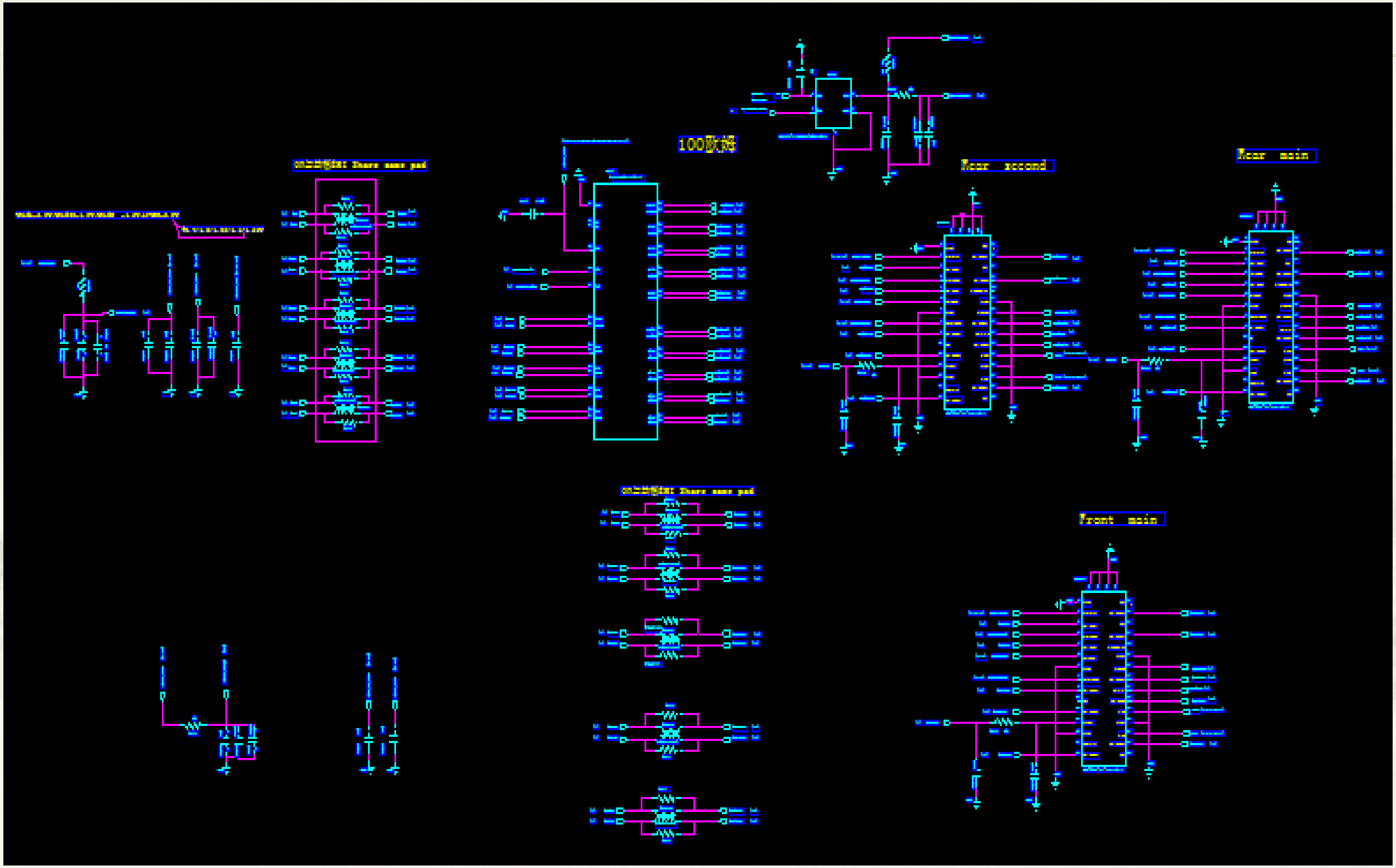
Speaker PA AW8737S U6701



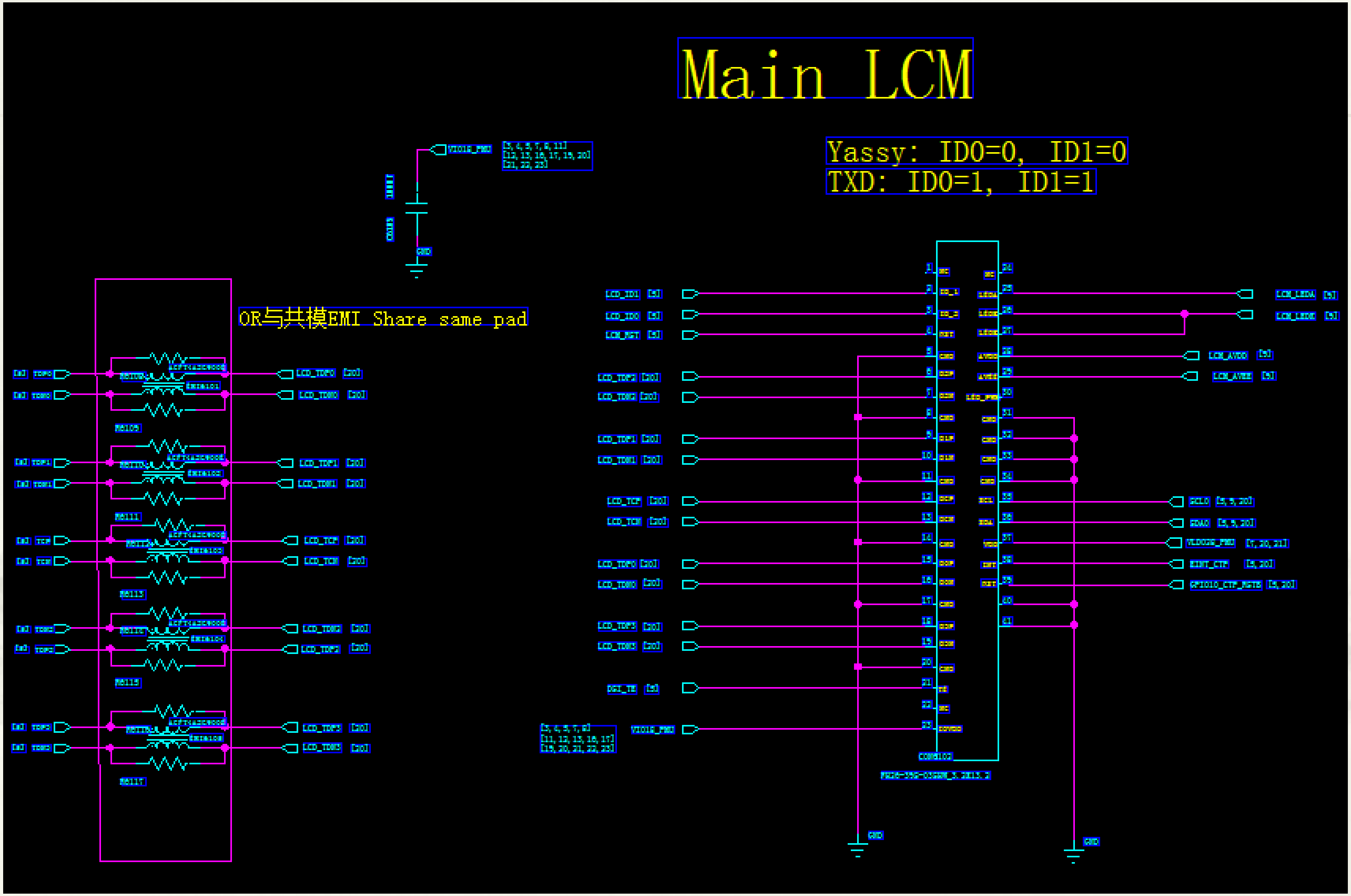
Receiver section



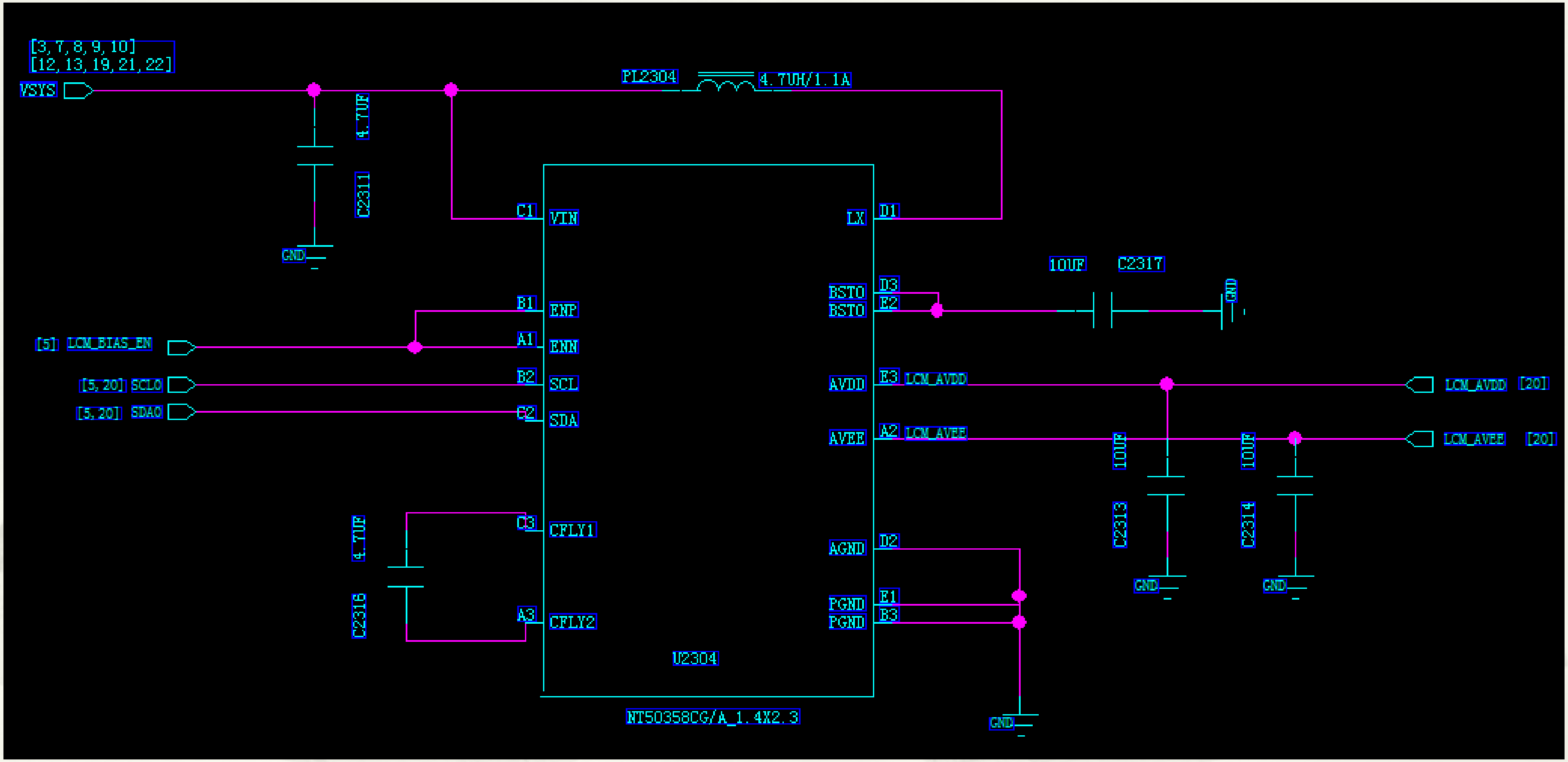
Camera section CON6210 & CON6201& CON6202



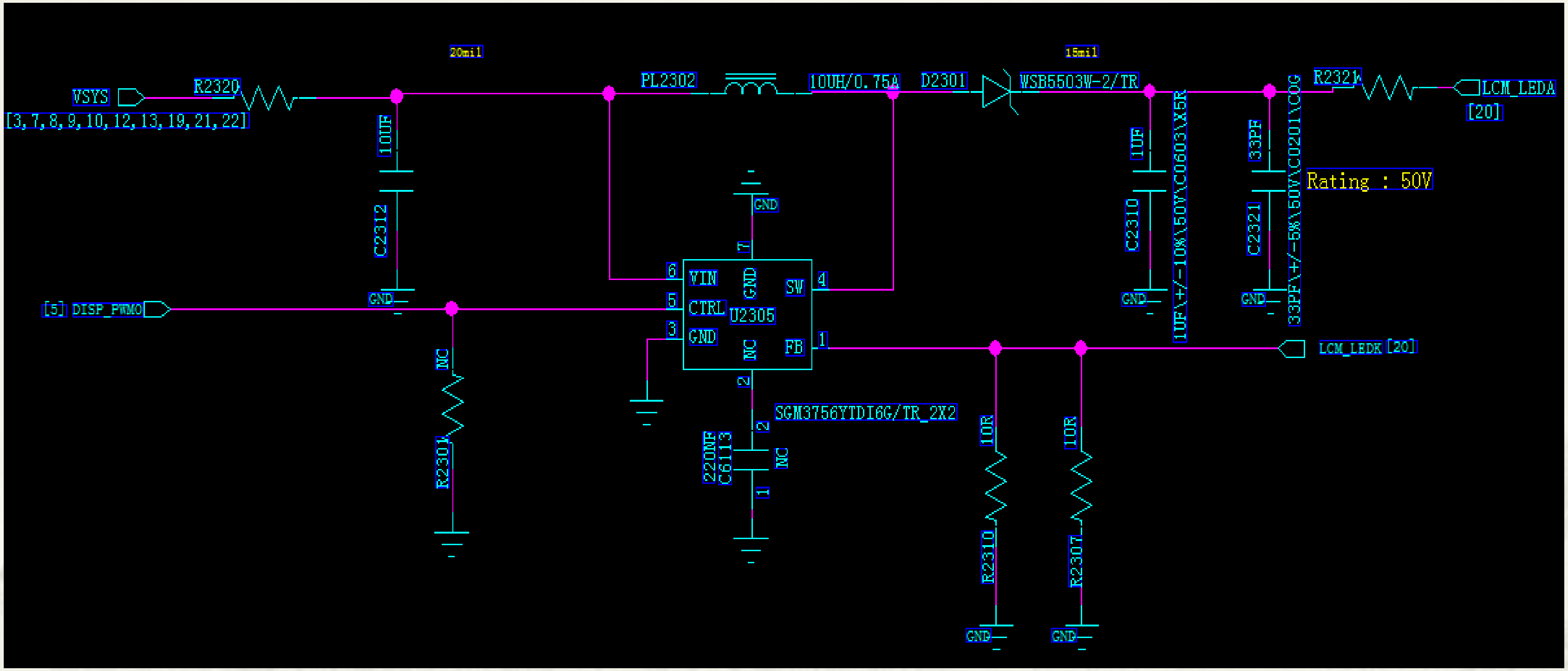
LCM section CON6102



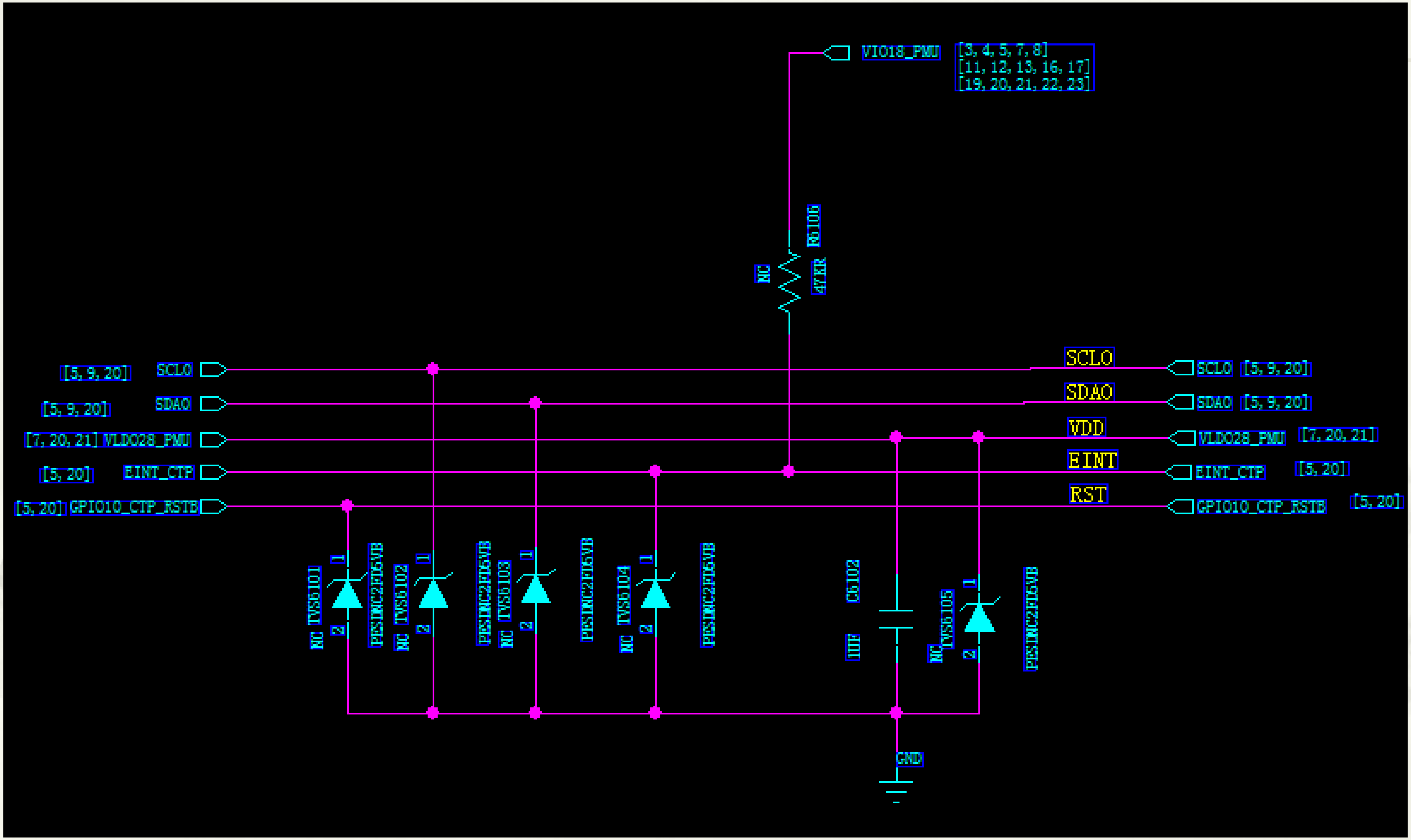
LCM Gate Drive U2304



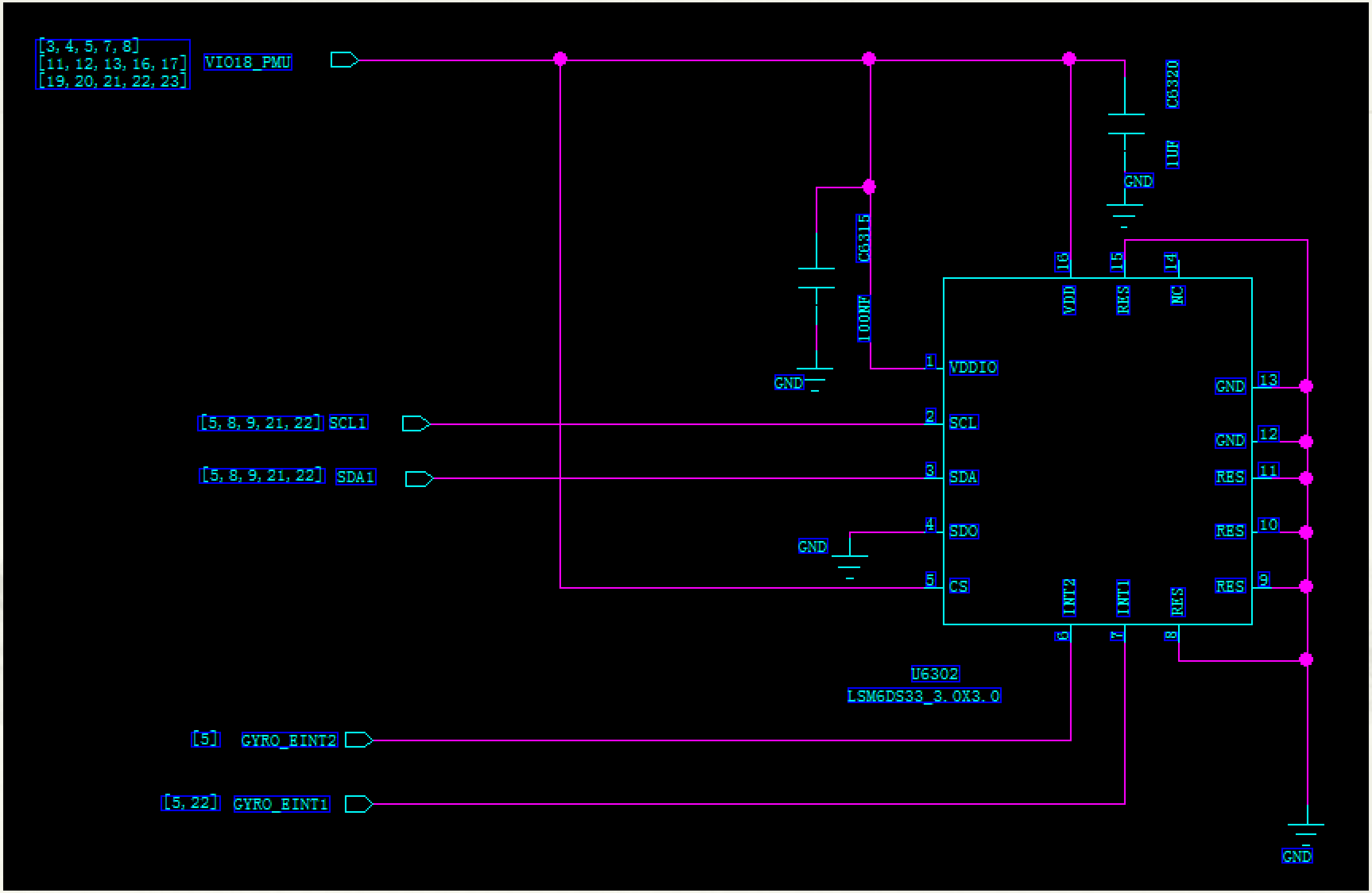
LCM Backlight LED Driver U2305



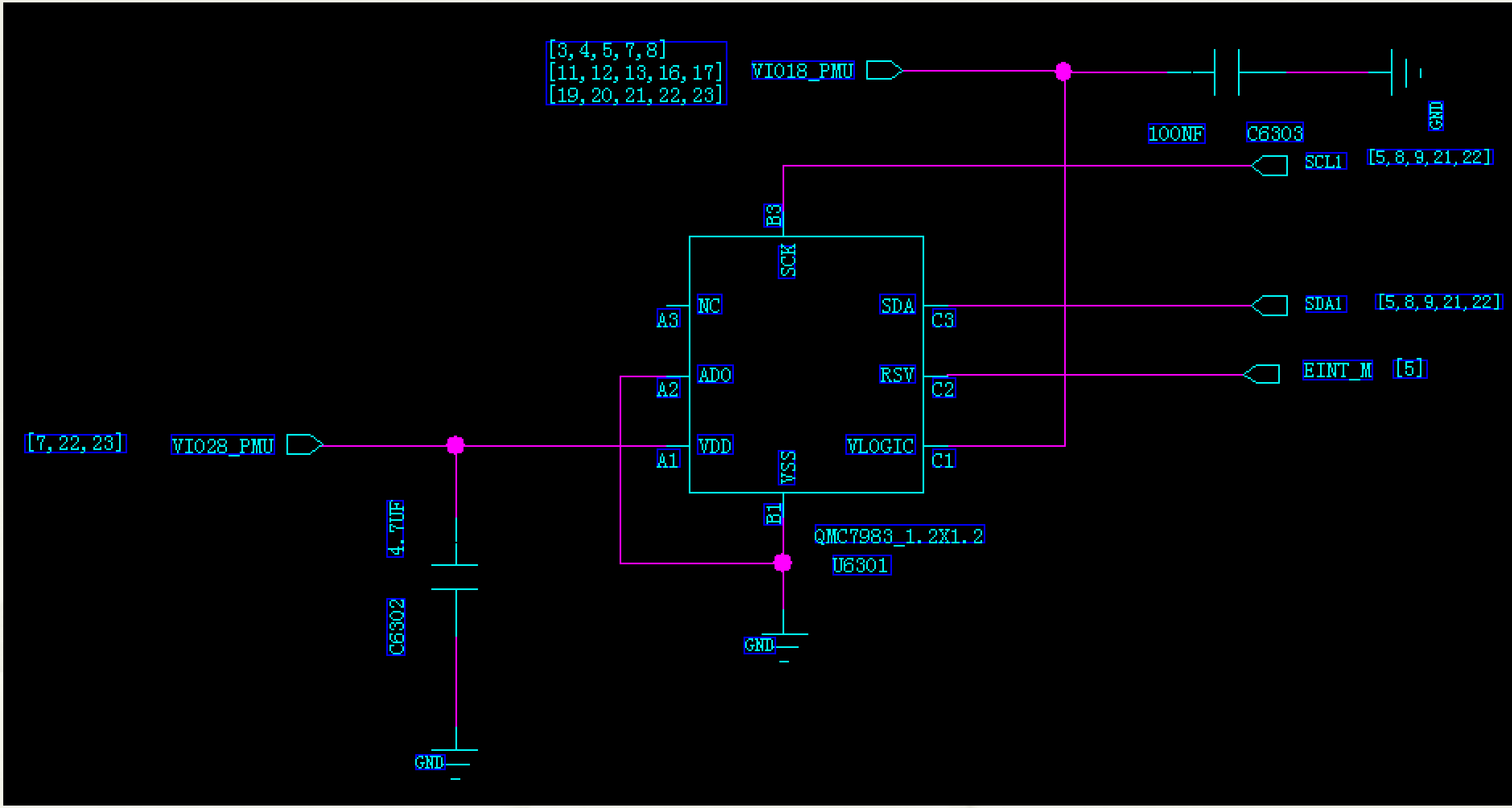
Touch panel section CON6102



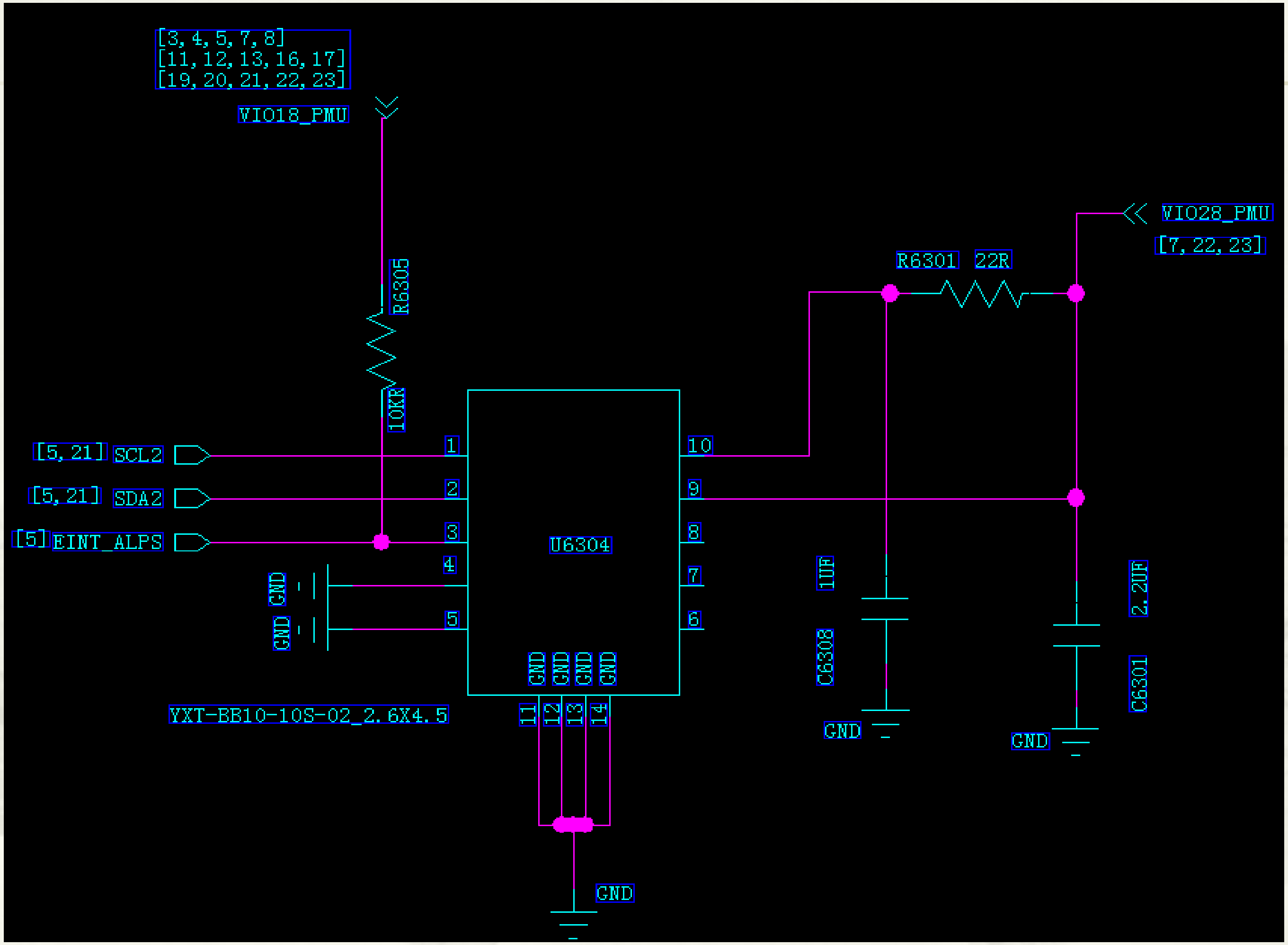
G-sensor section U6302



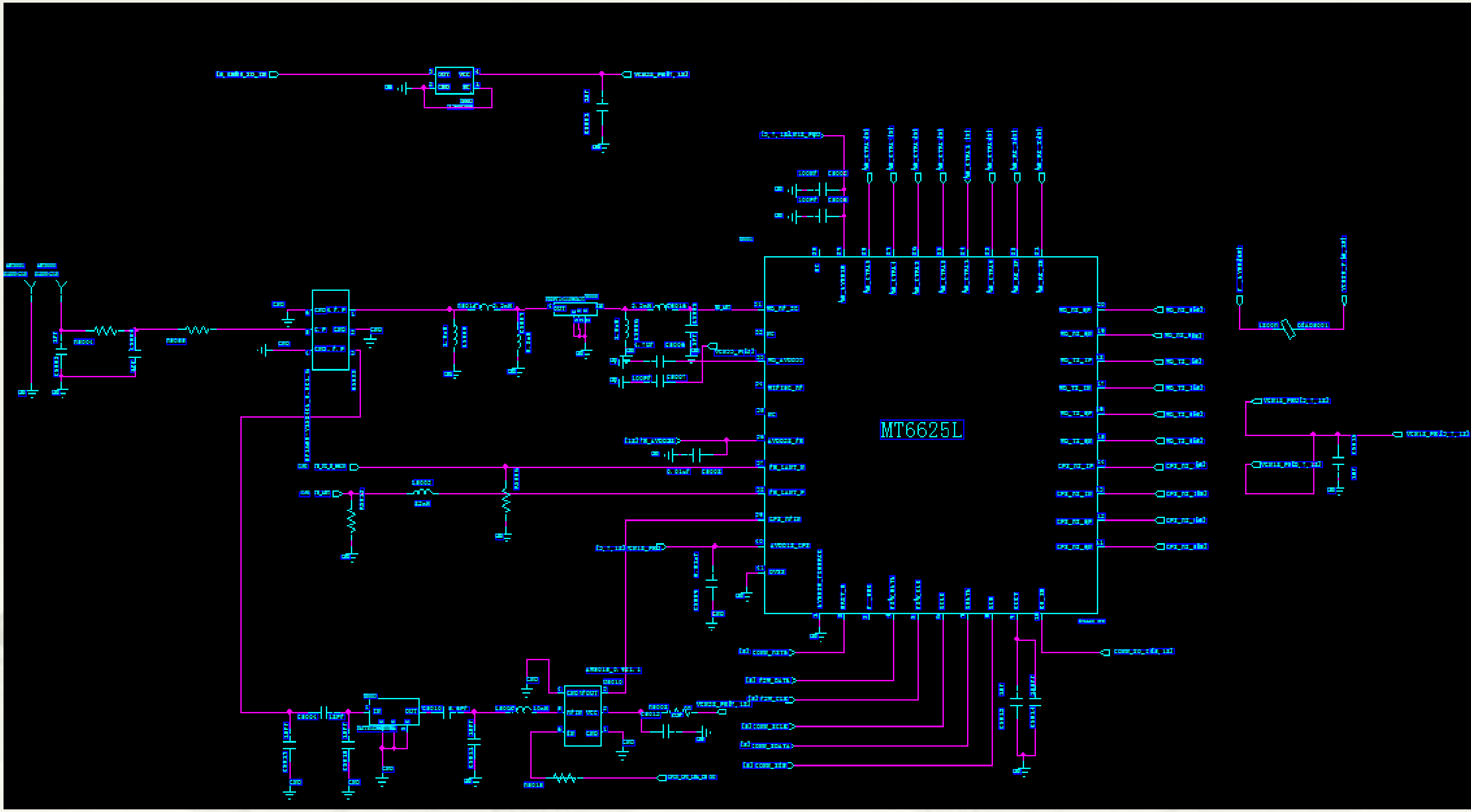
M-sensor section U6301



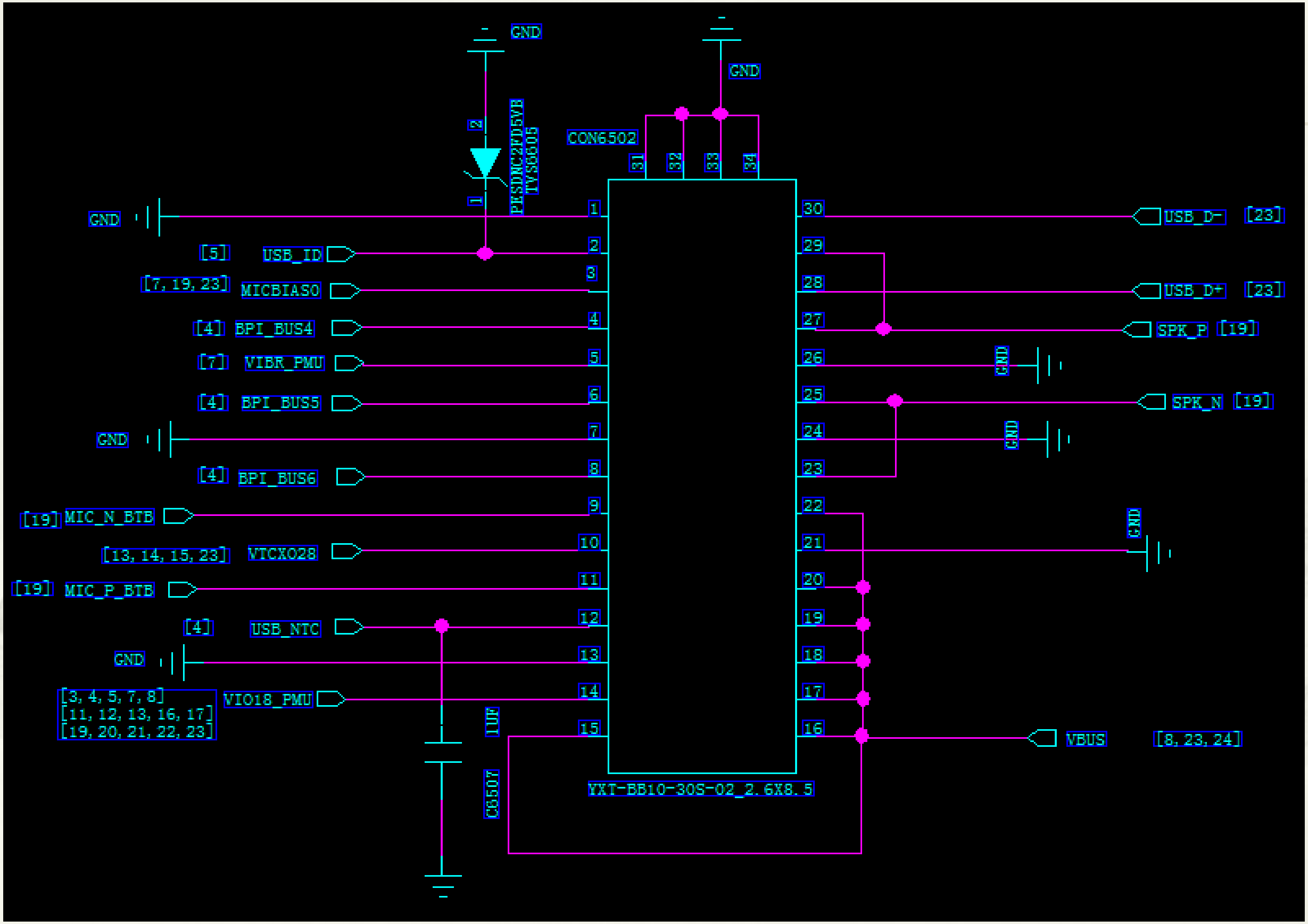
Ambient Light Sensor & Proximity Sensor section CON6304



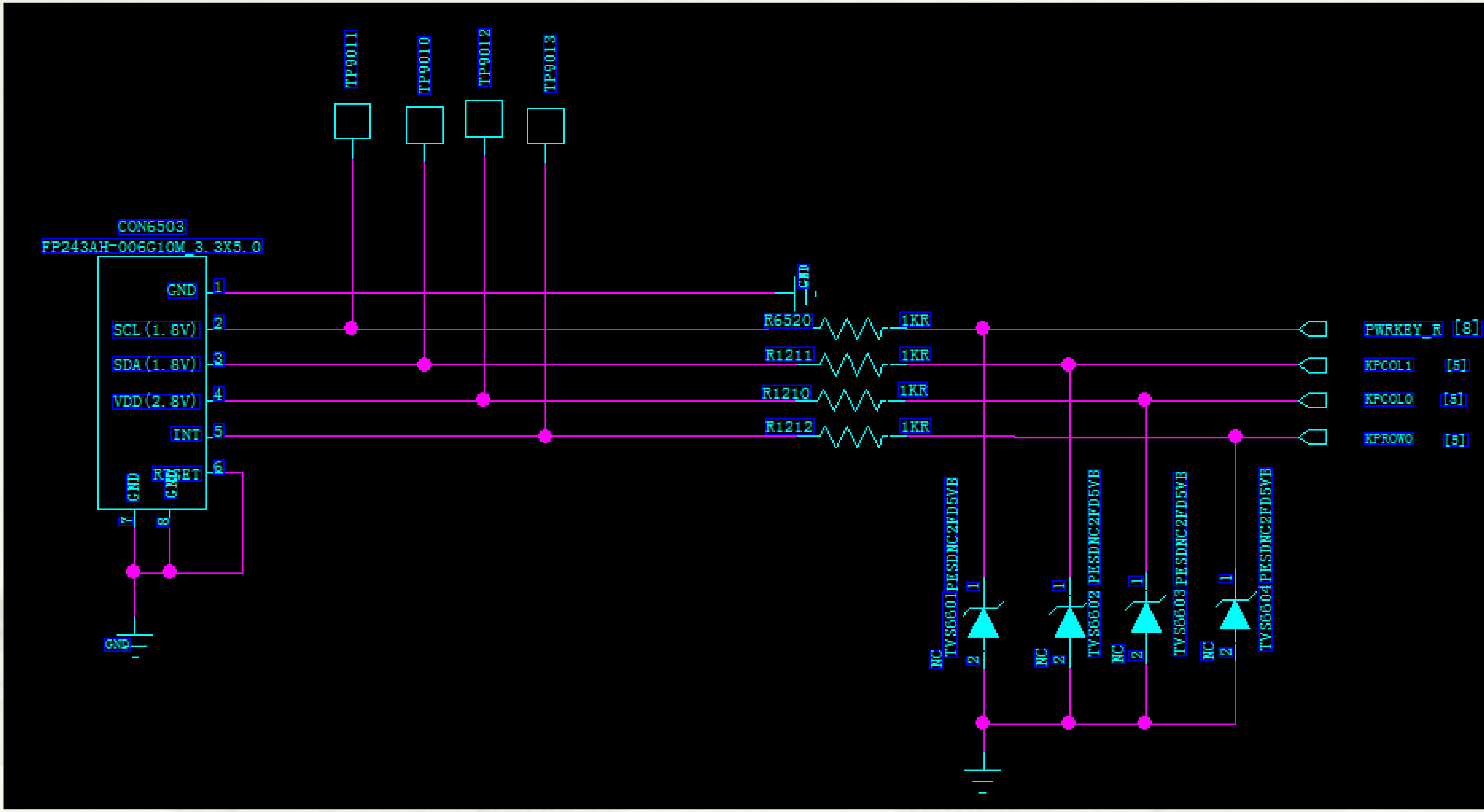
GPS/WIFI/FM/Bluetooth MT6625L U5001



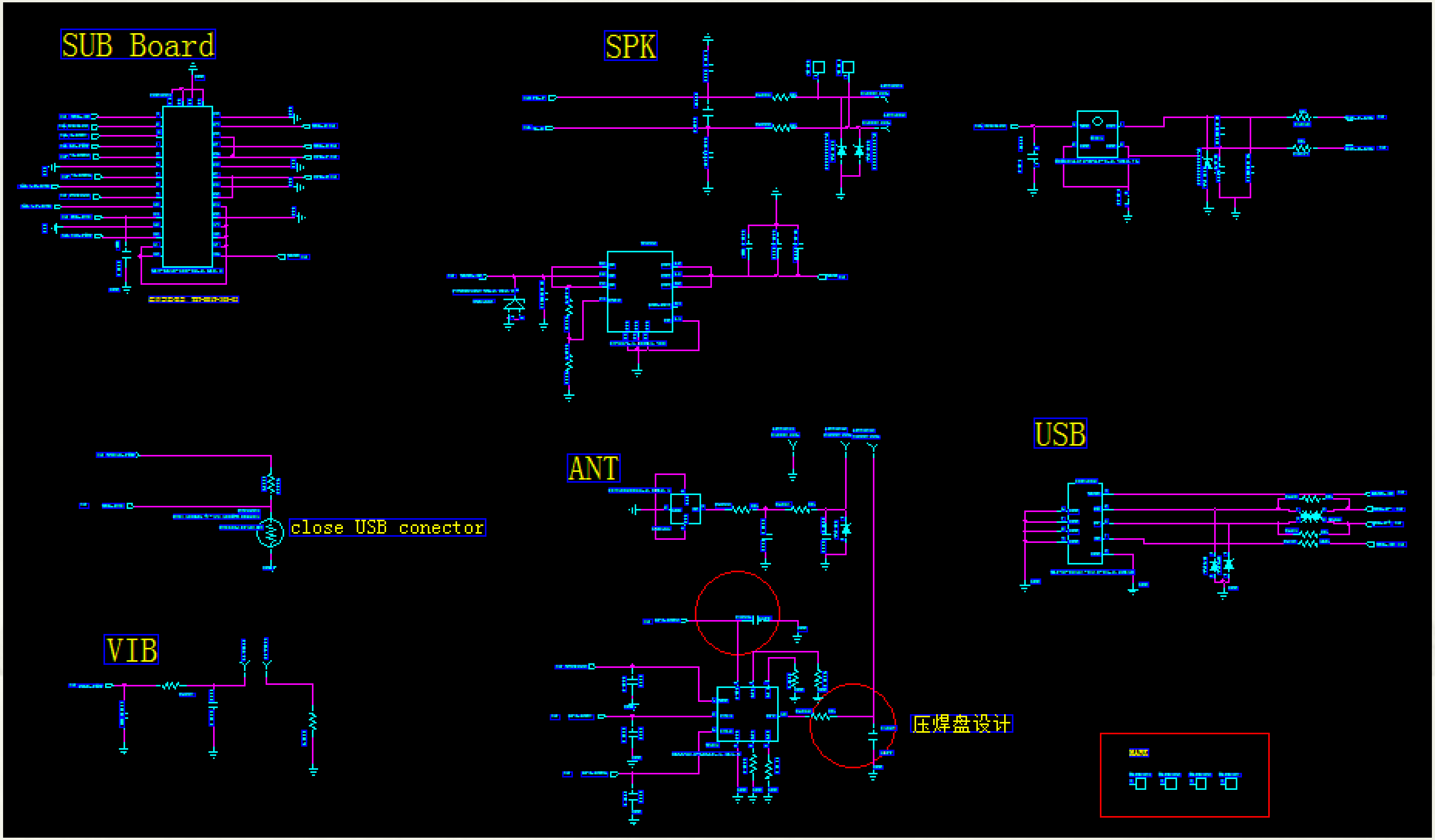
To sub board CON6602



Side key section CON6503



Sub board



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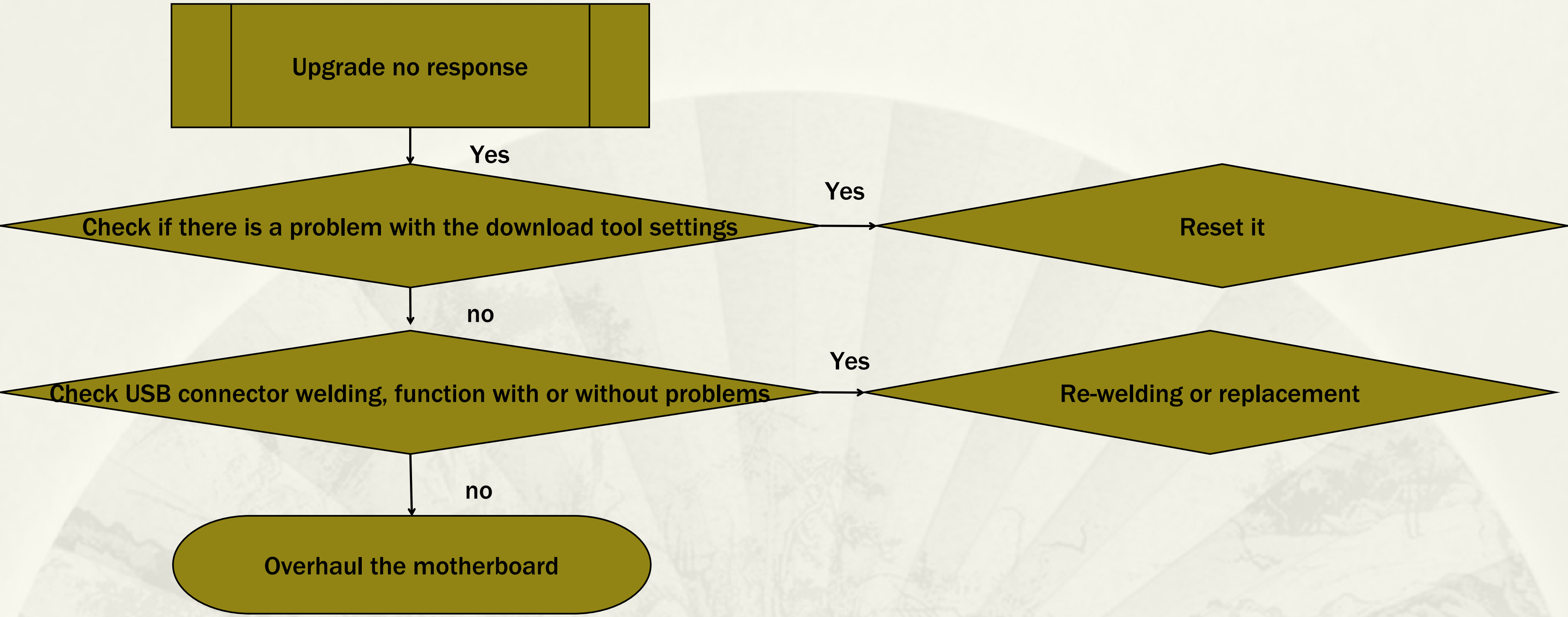
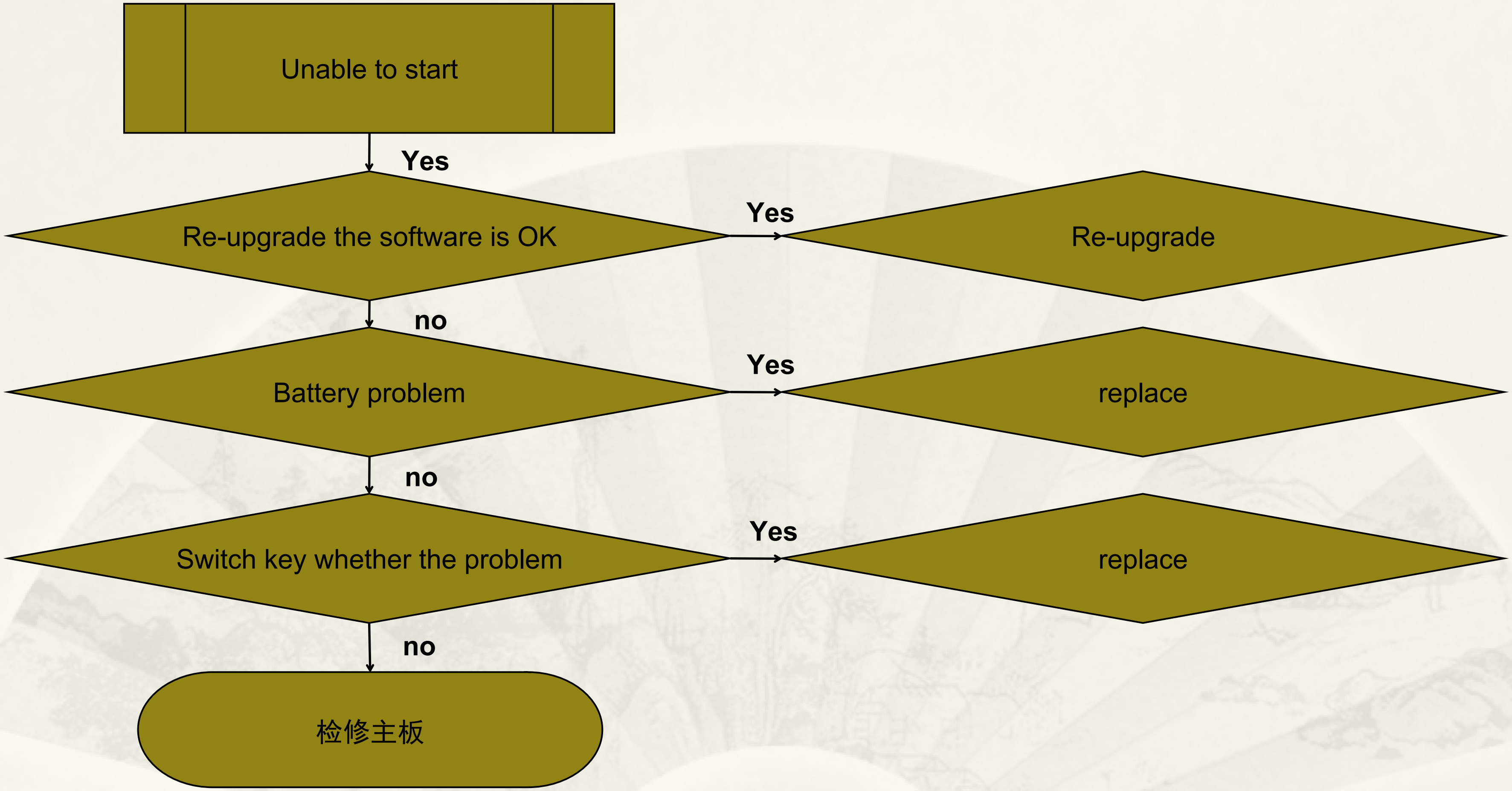
Required equipment and tools
ID appearance
Top view of the board
System Block Diagram
I2C address and power management
Important devices and key test points
Maintenance methods

1. Failure to fail analysis



2. Can not boot failure analysis



2.1 Can not boot failure analysis _ power voltage and current abnormalities

- With a multimeter or oscilloscope to measure the battery cell P + voltage is low, and the power is too low, the voltage is lower than 3.4V:
- - charge, to be greater than 3.6V voltage, re-start the boot action 。
- Battery P + voltage is greater than 3.6V, but the power leakage current ：

Make sure TVS2101 is abnormal ；

-- Check whether the charging circuit U2101 is abnormal ；

-- To confirm whether the radio frequency PA U3201, U3301 abnormal ；

-- To confirm U2305, U2301, U2304, U6002 whether the exception ；

-- Check PMIC U2001 for any abnormality ；

-- Check U6304 pin8 for short circuit ；

-- bove devices in addition to PMIC U2001 other than ，Other bit number removal does not affect boot ，Analysis can be used to exclude the removal method and then boot test 。

2.2 Can not boot failure analysis _ boot process current abnormal

- The VBAT voltage is high with a multimeter or oscilloscope, but the current is not normal during startup

--To confirm whether there is a chip fever

--Check that the PMIC voltage output is normal

--Make sure the CPU is abnormal

2.3 Can not boot failure analysis clock

* Measure the VBAT voltage with a multimeter or oscilloscope is high

--Make sure that the output voltage of PMIC U2101 is abnormal, There is an exception is welding or replacement PMIC further confirmed, if no exception is the next step

--Confirm 32.768K clock X2101 output 32.768K clock is normal

--Confirm 26Mhz clock U3103 output 26M clock is normal

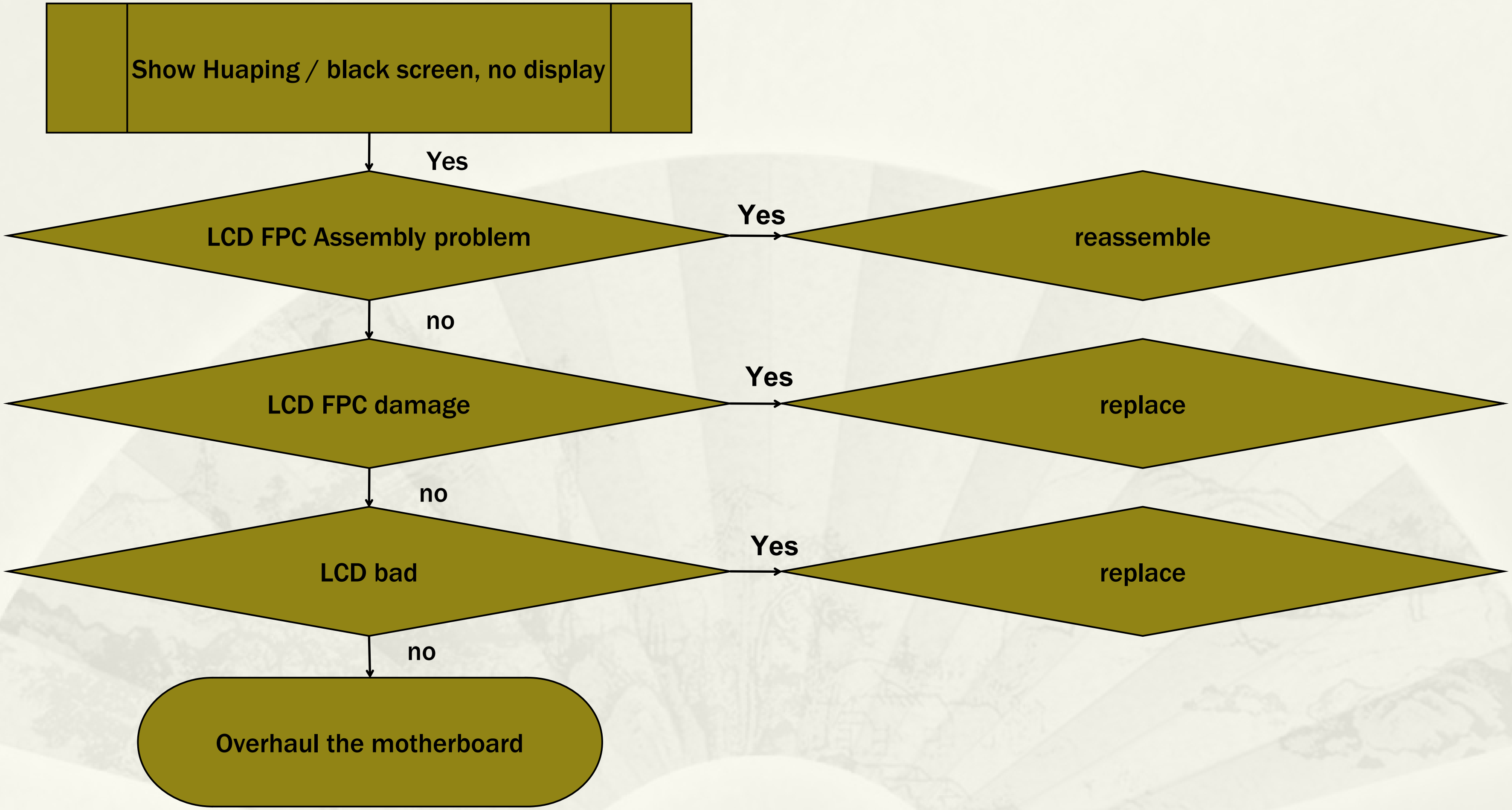
--To confirm whether the RF transceiver chip U3101 abnormal, because the CPU to the 26M clock through the U3101

2.4 Can not boot failure analysis eMMC

-- Confirm eMMC monomer U4102

-- Confirm R4103，R4104

3. Display abnormal failure analysis



3.1 Display abnormal failure analysis LCM

-- Make sure CON6102 is empty or missing 。

--Has confirmed the completion of the boot, the measurement power supply VIO28_PMU and VIO18_PMU is normal

--Check whether the backlight is normal light, if the backlight does not need to check the backlight drive chip U2305 and its peripheral devices are normal. Backlight is the next step

--Check that the bias chip U2304 is working properly

-- Measure whether all the signals on the R6108-R6117 are signaled, and if not all signals, confirm that the CPU U1001 looks and X-ray, there is an abnormally solderable or further replacement

3.2Display Abnormal Fault Analysis Backlight Chip

-- Make sure U2305 is empty or missing 。

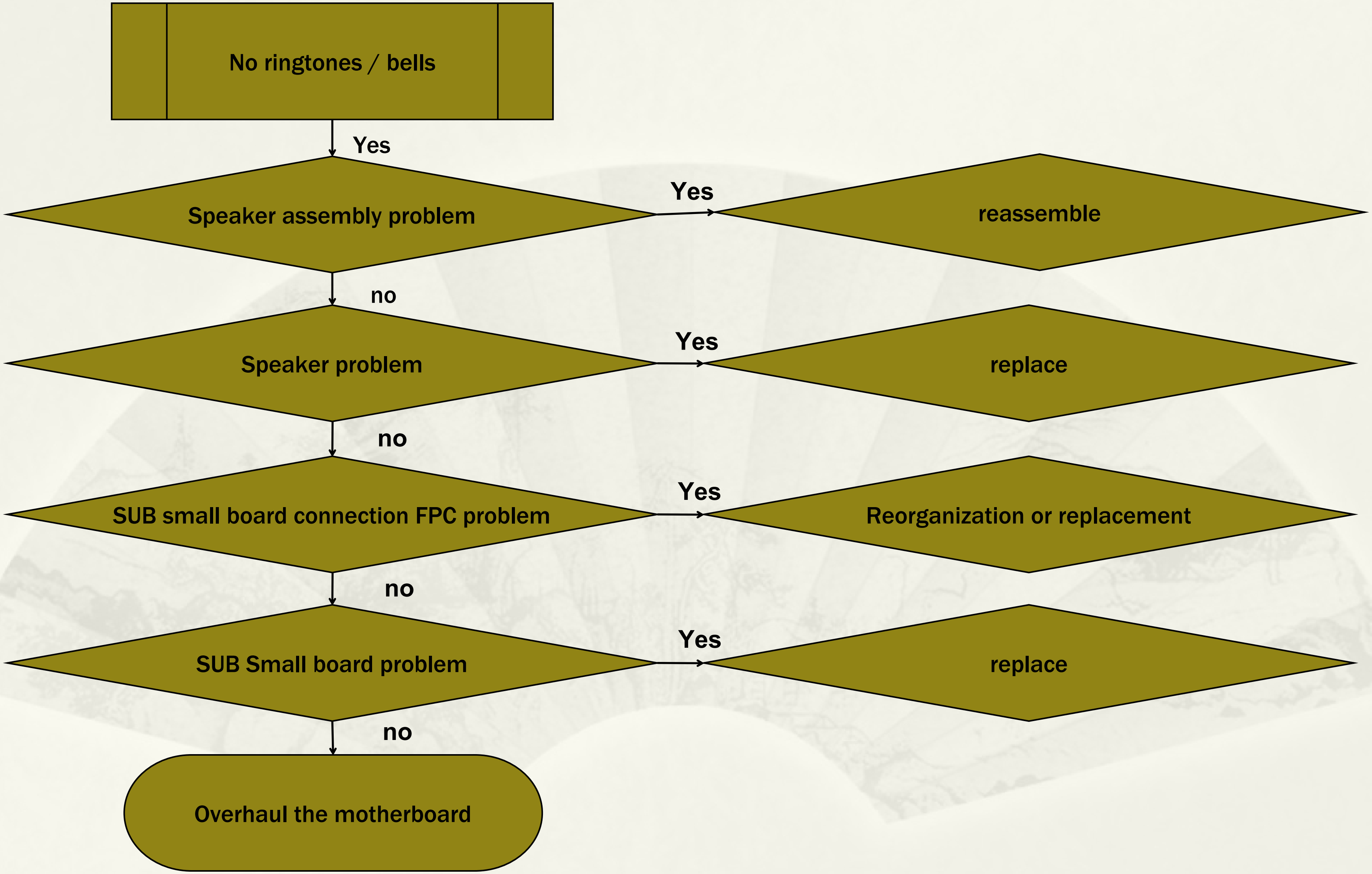
-- Check if the input voltage VBAT is normal

--Check whether the FB signal of U2305 has an output voltage of about 0.2V

-- confirm PL2302、D2301、R2310、R2307、C2310、C2321、R2312，exception is re-welded or replaced

-- Has confirmed the completion of the boot, the measurement power supply LCM_LEDA there is no about 19V level, the output voltage is not normal replacement U2305

4. Ringtones fault analysis



4.1Ringtones fault analysis

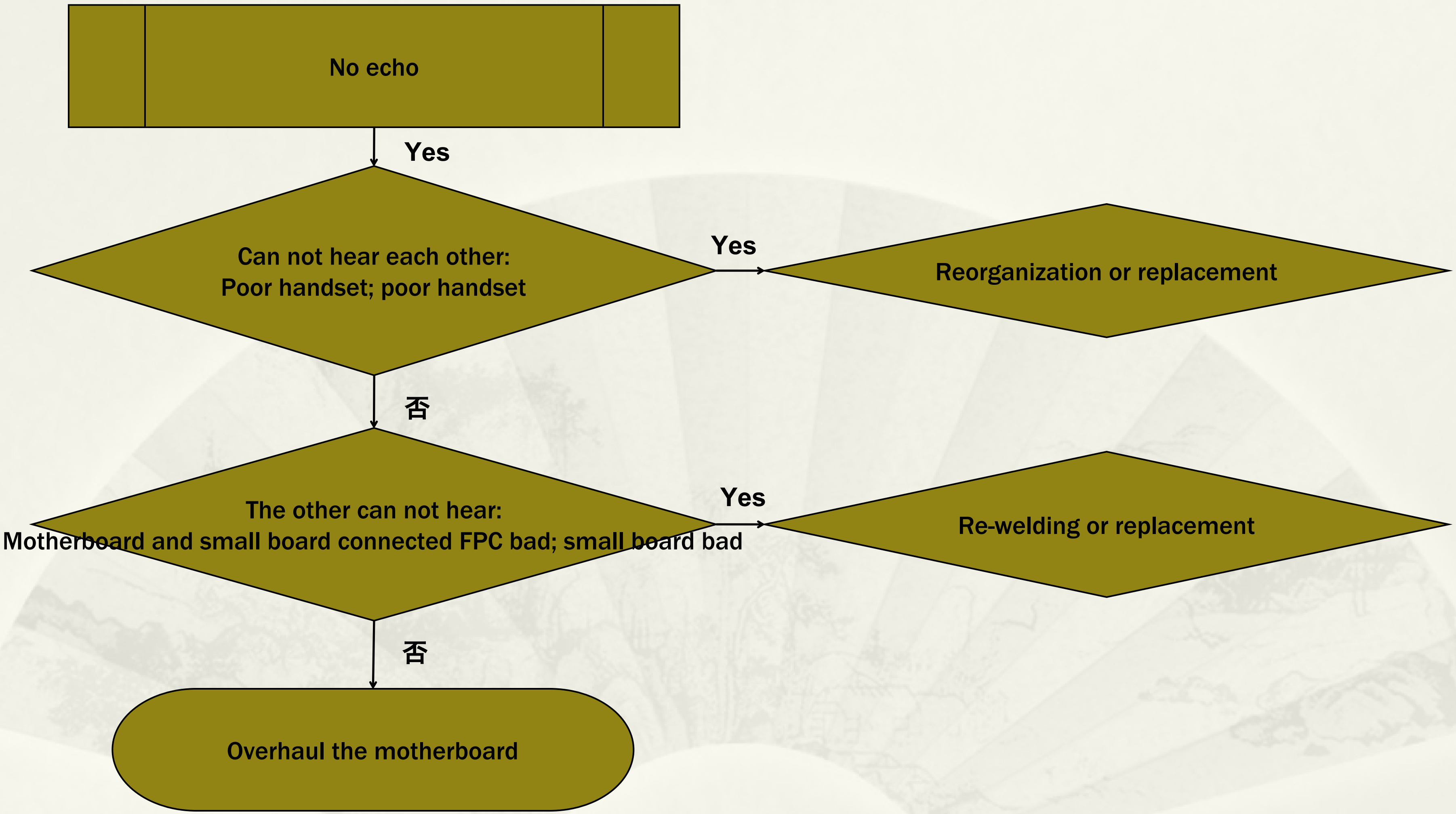
-- Check if the audio amplifier U6002 has an empty or missing part 。

-- Confirm C6790 , C6705 , C6707 , C6703, C6709, C6701, C6702, C6704, R6704, R6703, R6705, R6001, R6003 are normal, abnormal re-welding or replacement

--Make sure that the main PCB connector CON6502 has an empty weld or a loss item

-- Confirm PMIC U2001 appearance and X-ray, no exception can be welded or further replacement

5. No echo failure analysis



5.1 No echo failure analysis

-- Check the peripheral material of the handset C6032, C6025, C6031, whether there is any welding or missing parts 。

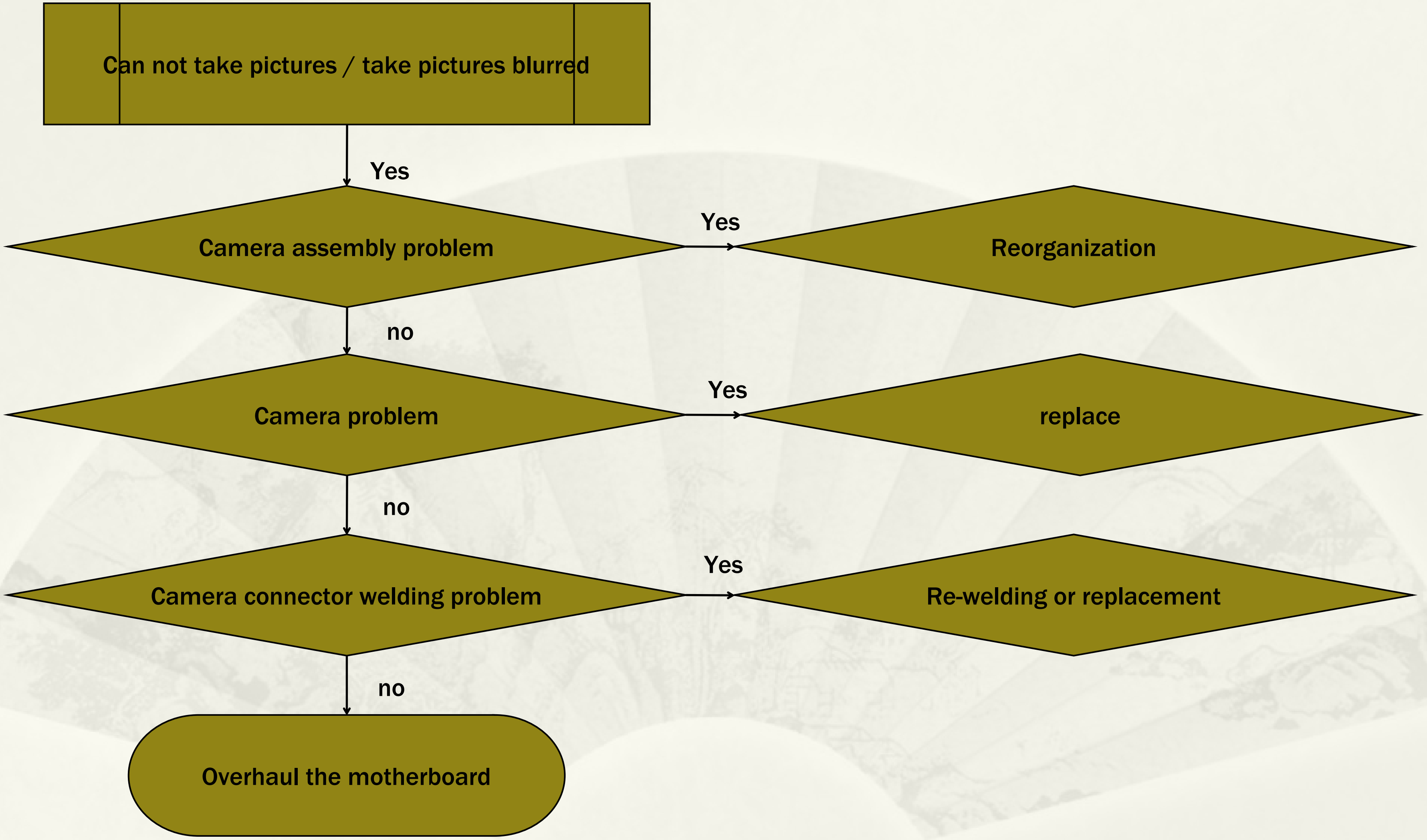
-- Check whether the motherboard main MIC material R6027, R6028, C6029, C6030, C6027, C6033 have empty welding or missing parts 。

-- Make sure that the main and subplane connectors CON6502 PIN19-23 have Weld and the connector is damaged 。

--To confirm the small board on the main microphone material C101, MIC1, TVS1006, C1014, C1015, C1016 , C1018 , C1019 is normal

--Confirm PMIC U2001 appearance and X-ray, no exception can be welded or further replacement

6. Poor camera failure analysis



6.1Poor camera failure analysis

-- Press the subject and the sub-picture to confirm whether the main connector CON6201, CON6210 and the sub-connector CON6202 are empty or missing 。

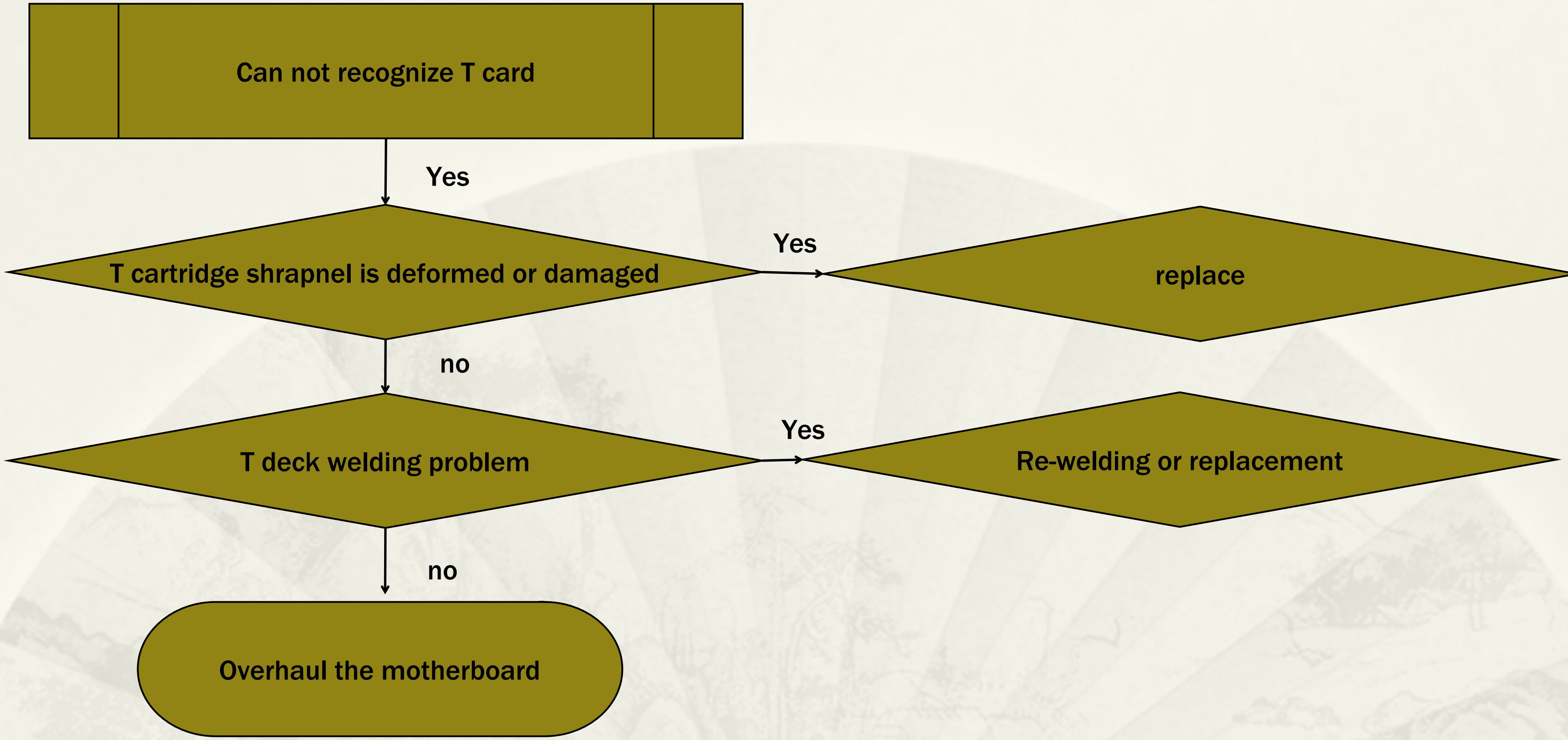
-- Make sure the power supply is normal 。

-- Check that the mains short circuit module R6220-R6229 is normal 。

Make sure that the sub-camera short circuit module R6207-R6219 is normal 。

--Confirm the appearance of CPU U1001 and X-ray, no exception can be welded or further replacement

7.T card failure analysis



7.1 Fault Analysis of T Card

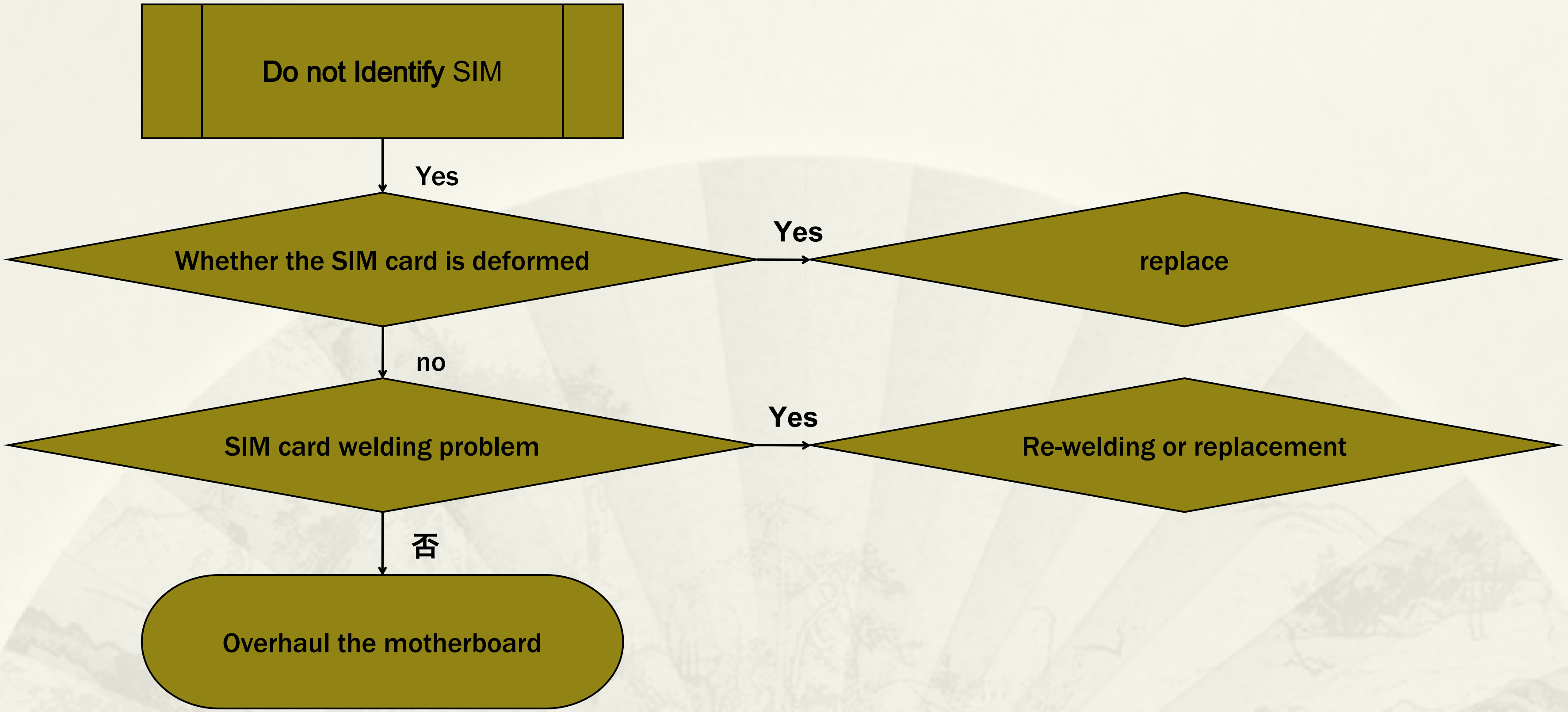
-- Check if the relevant TVS tube has an empty weld or a loss item 。

-- Check whether the power supply is normal, VMCH_PMU output is 3.0V, confirm C4003 。

-- Check that the interrupt EINT_SD is normal, CON4001 and peripheral devices 。

--Confirm the appearance of CPU U1001 and X-ray, no exception can be welded or further replacement

8. Do not Identify SIM failure analysis



8.1 Do not Identify SIM failure analysis

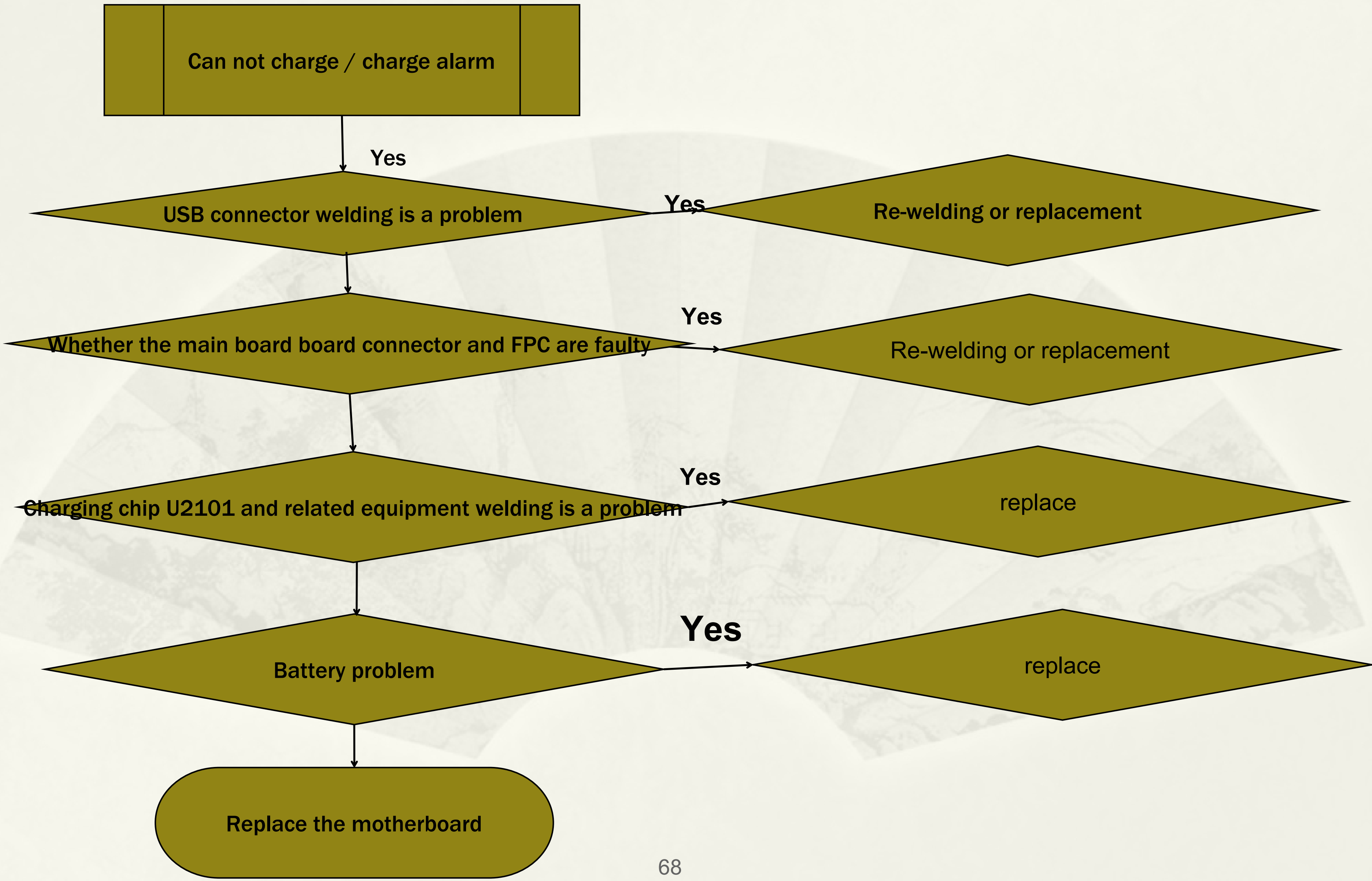
--Make sure that the SIM card connector CON4002, CON4003 is defective, empty, and damaged

-- To confirm whether the power supply is normal, VSIM1_PMU, VSIM2_PMU output is normal 。

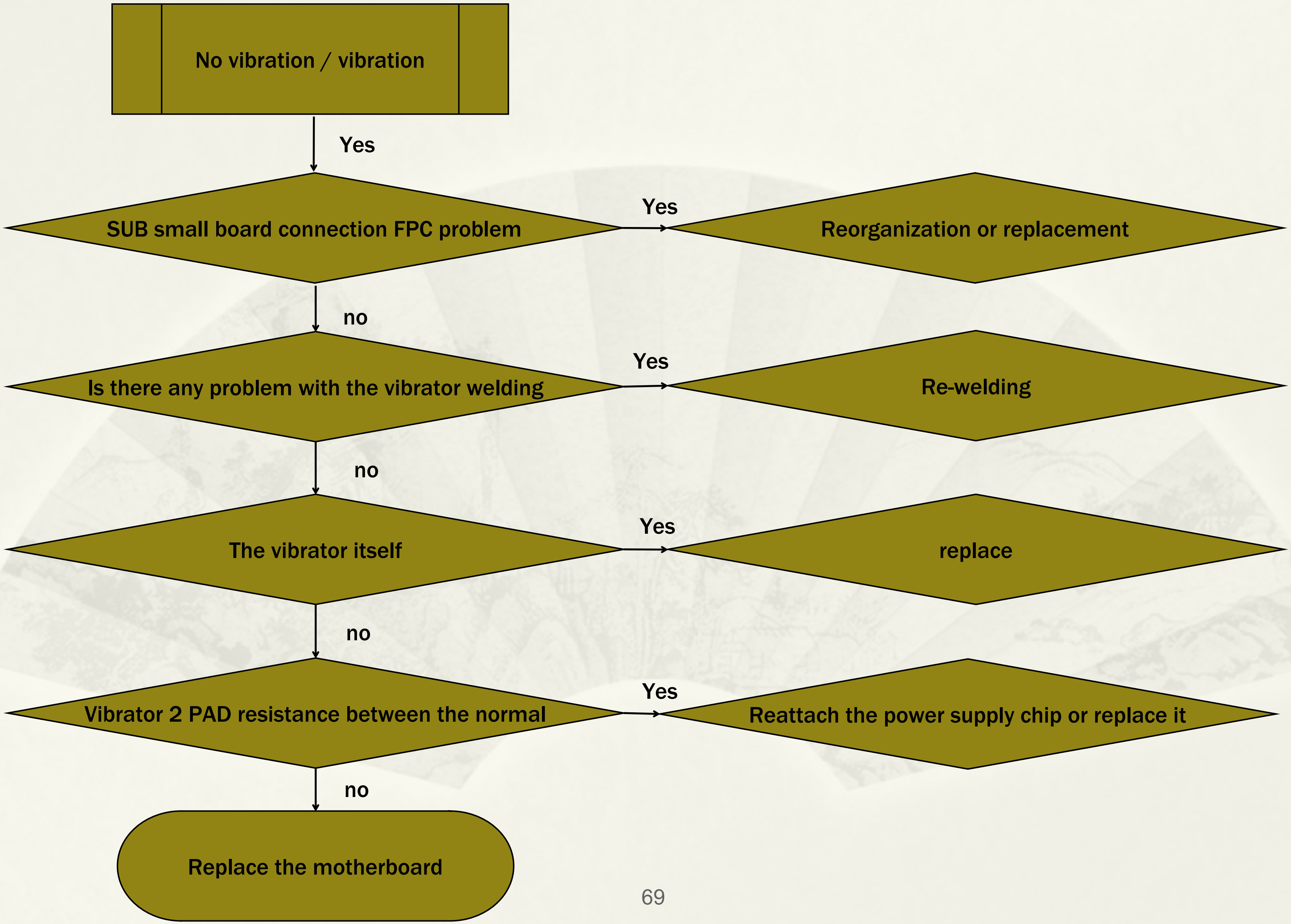
-- Confirm interrupt INT_SIM2, INT_SIM1 is normal, confirm R4002 , R4003 , C4008 , C4002 。

--Confirm the appearance of CPU U1001 and X-ray, no exception can be welded

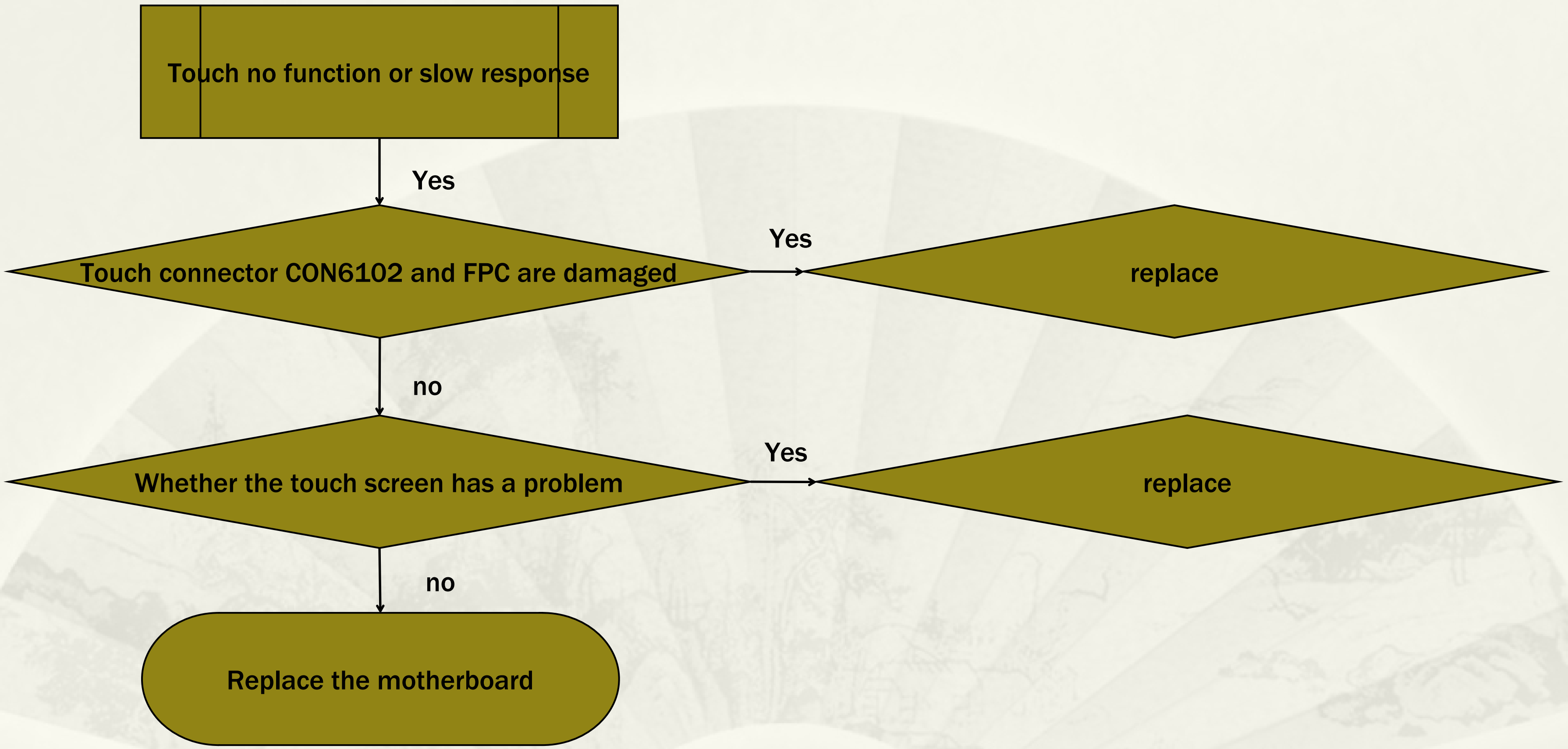
9. Charge failure analysis



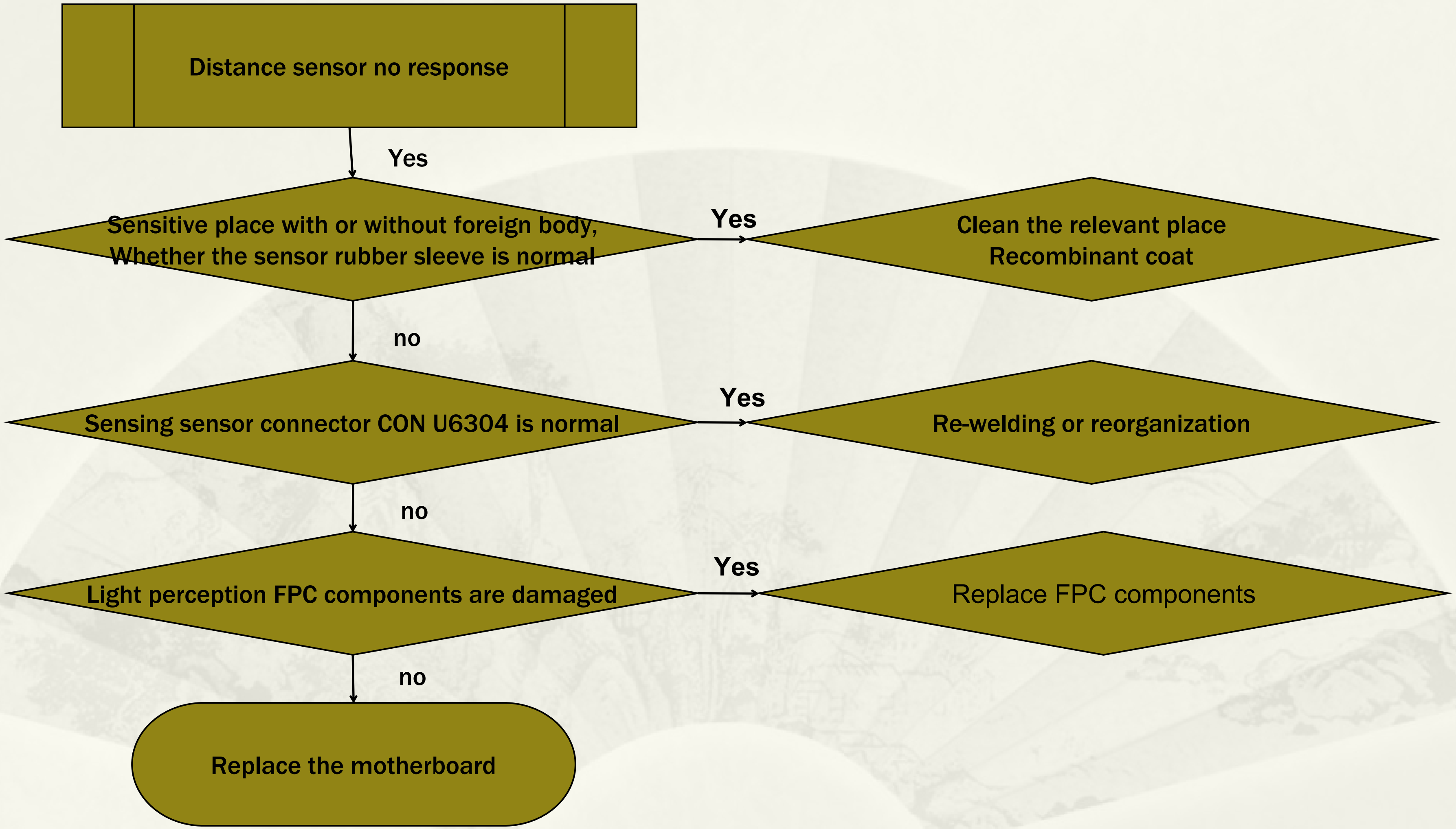
10. Vibration failure analysis



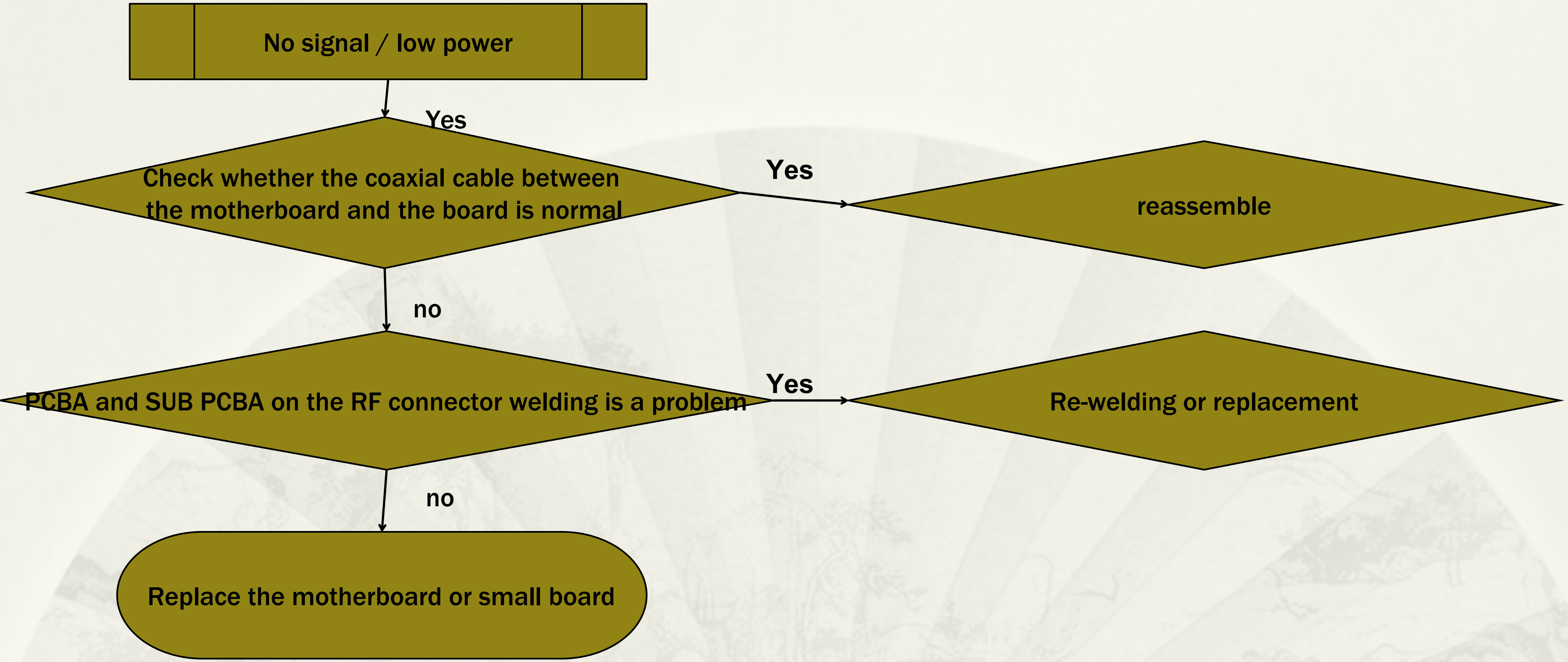
11. Failure analysis of touch screen



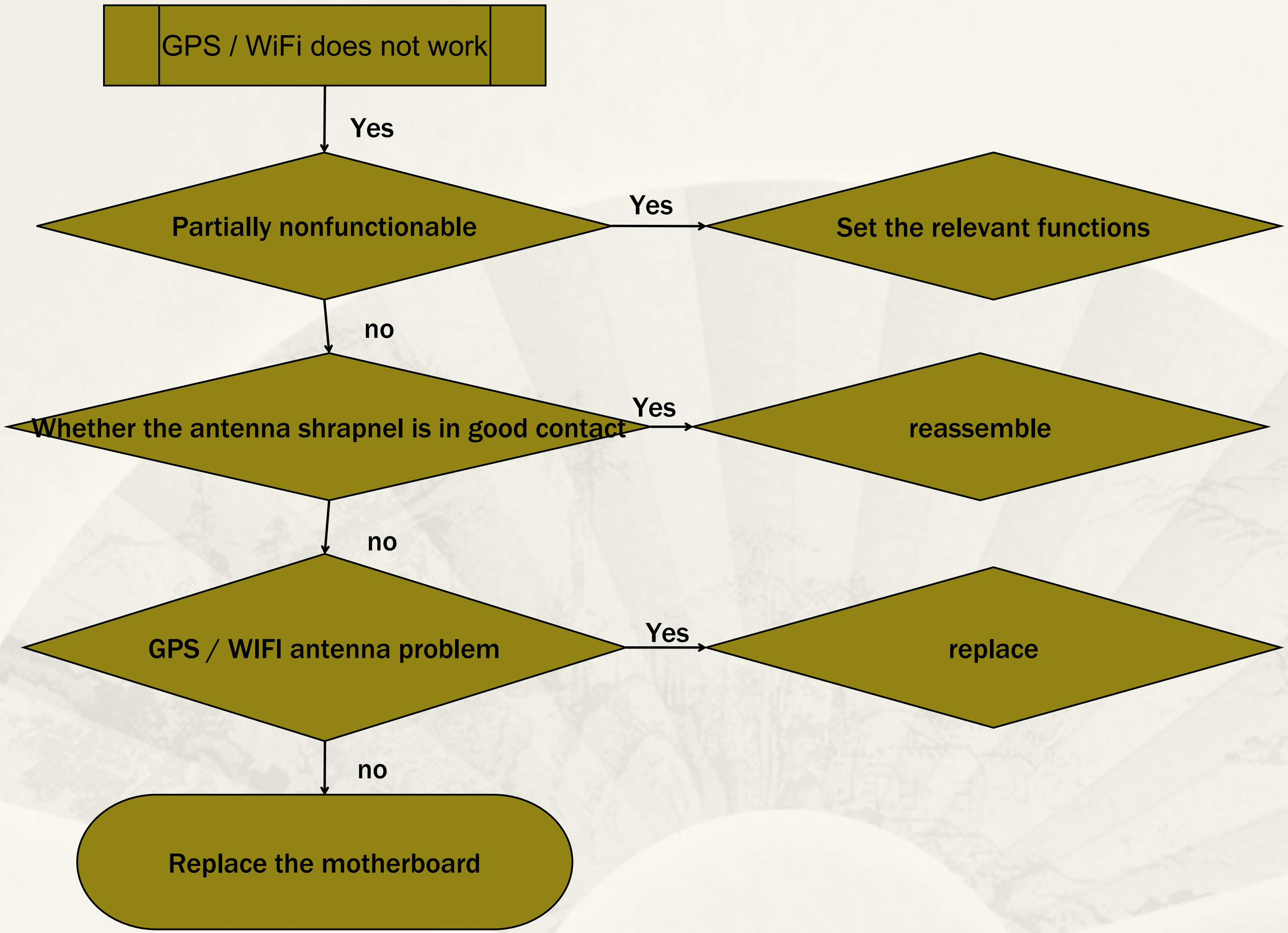
12. Analysis of Distance Induction Faults



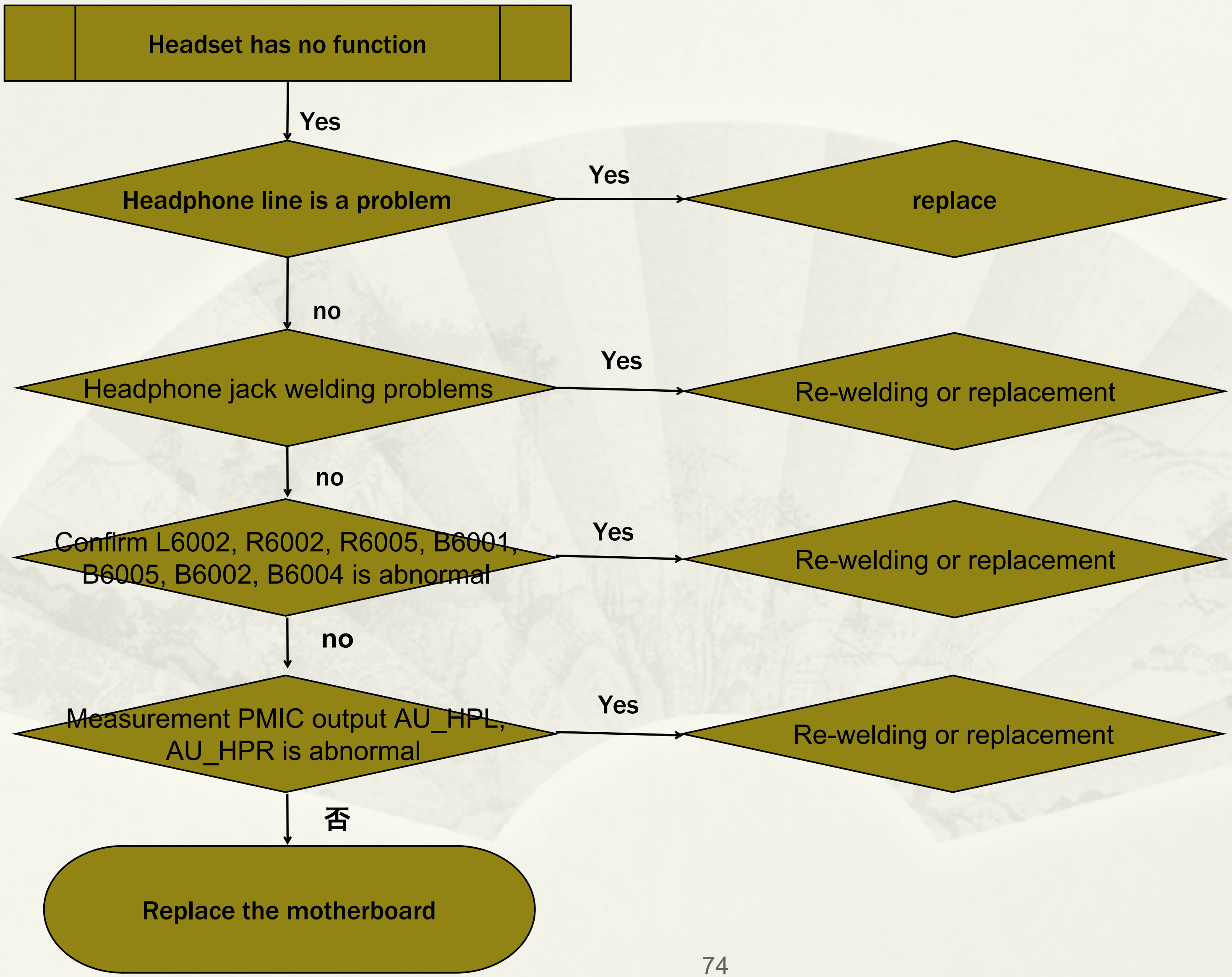
13. No signal failure analysis



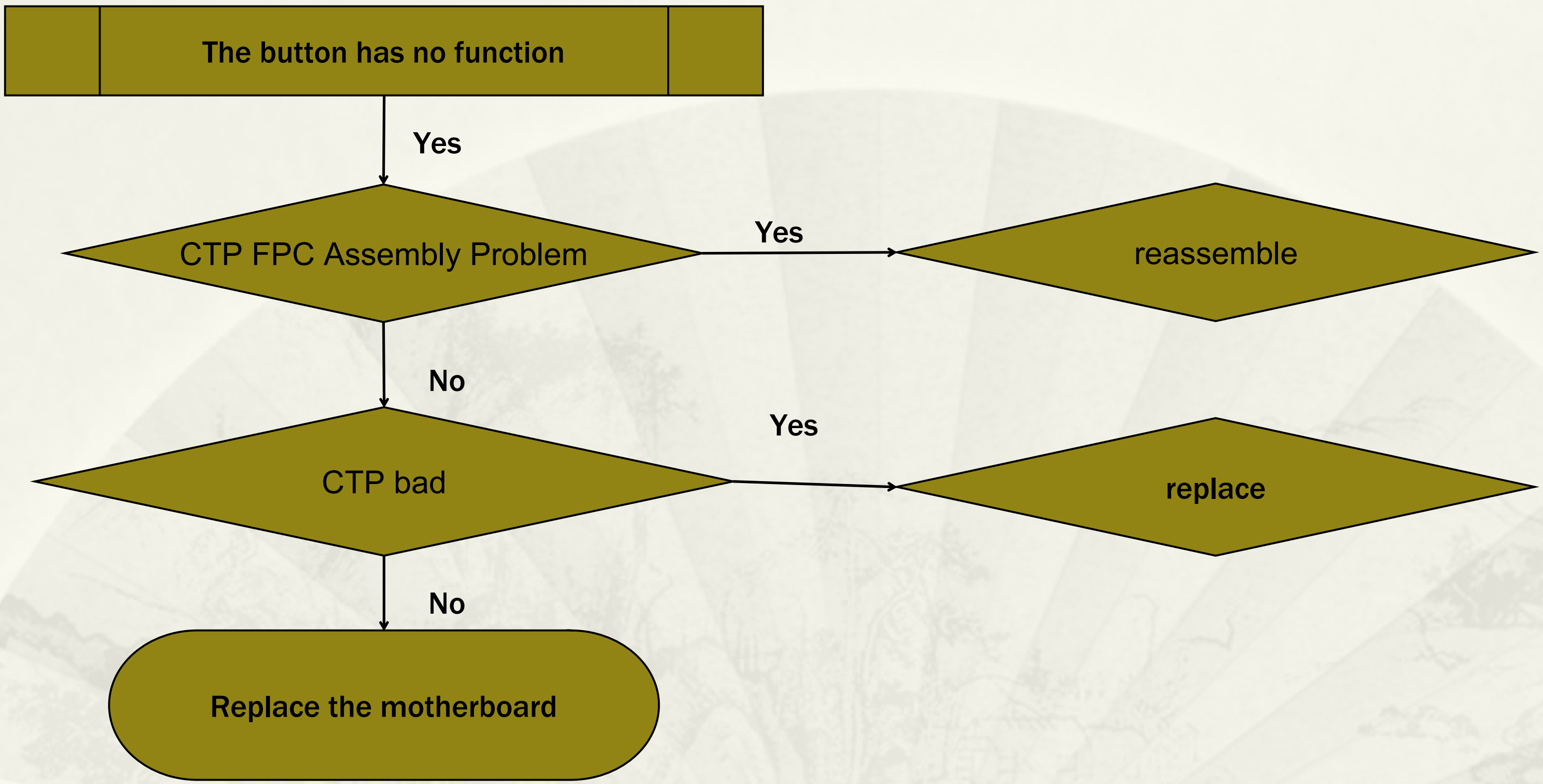
14. GPS / WiFi fault analysis



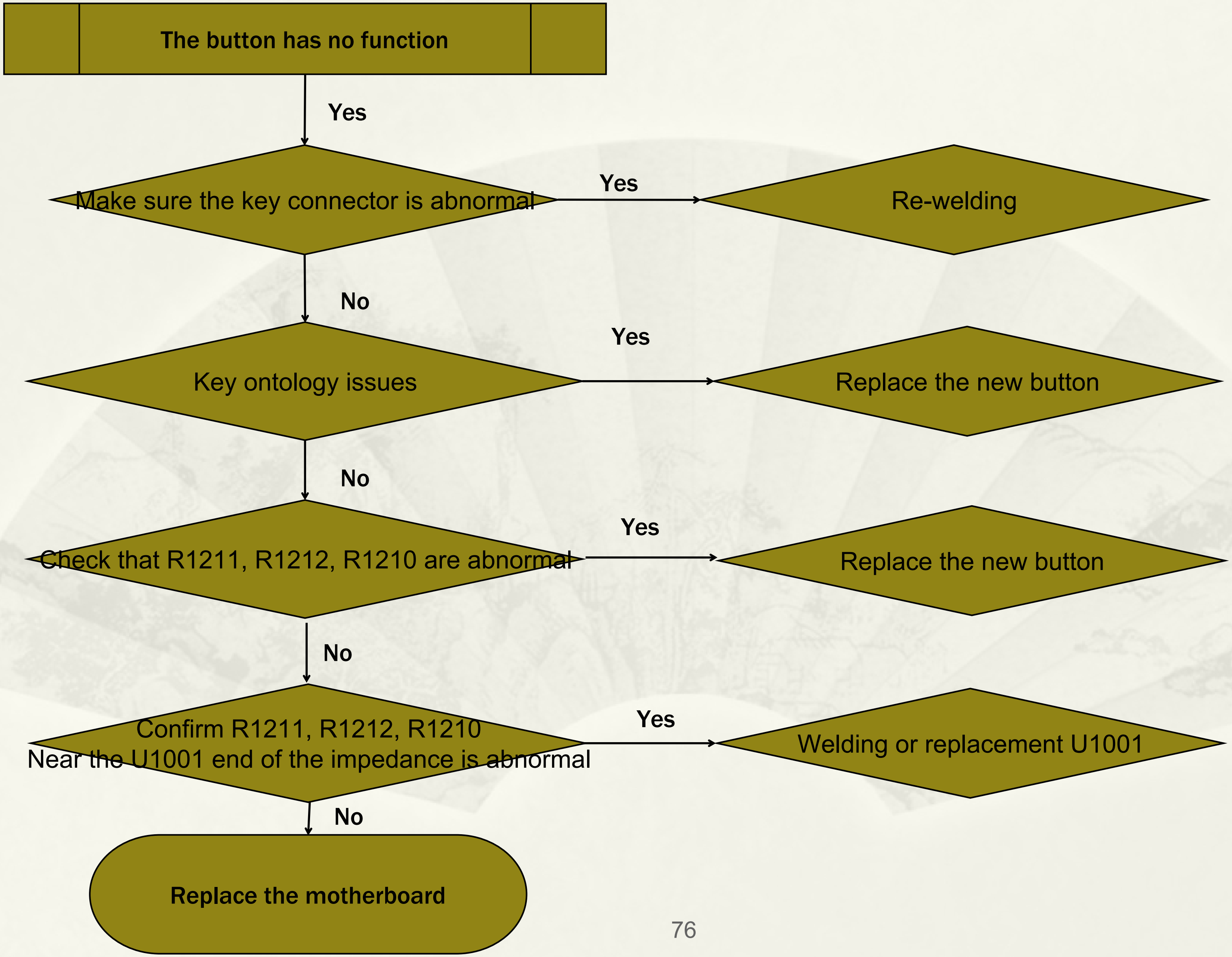
15. Headset problem analysis



16. Virtual Key Failure Analysis



17. Key failure analysis



Thank you ！